Dissociating Kinematics, Force, and Control in Video World Models for Manipulation

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Abstract

Project-status note (2026-04-30). This document is no longer a paper draft. It is a working record of the experimental state after the 2026-04-23 methodology reset. All previous abstract content was based on the episode-mean probing pipeline (`archive_old_wrong_probe/`) and is invalidated. The current artifacts are layer-wise probing results on the recollected PhysProbe dataset (12,100 episodes, valid contact GT) using V-JEPA 2 ViT-L raw residual features, plus a battery of latent-trajectory analyses. Sections that follow record *Setup / Outputs / Key observations / Known limitations / Status* per experiment block; no narrative or claim is intended.

Note 1 — Evidence inventory

Status. Replaces the prior introduction. All earlier claims (objective-specific dissociation, PEZ peak layers in the 11–12 range, native contact-force ranking V-JEPA 2 $>$ VideoMAE-L $>$ DINOv2-L, action-chunk control results) were produced by the invalidated episode-mean pipeline. They are not relied on here. The current valid evidence is restricted to V-JEPA 2 ViT-L (no VideoMAE / DINOv2 reruns yet), Variant A and Variant B feature pools, and the trajectory-analysis battery on the recollected dataset.

What we have right now.

- Layer-wise per-target Ridge probing across all six tasks under *Variant A* (1024-d spatiotemporal mean) and *Variant B* (8192-d, 8 temporal slots $\times$ 1024-d spatial mean).
- Extended target set per task: kinematic, rotational, contact, and progress-related variables (e.g. `insertion_depth`, `drawer_opening_extent`, `axial_progress`).
- Trajectory analysis on Variant A: per-layer intrinsic dimension, path/curvature/direct-distance ratios, cross-layer CKA, cross-task CKA, partial RSA controlling for pos+vel, tangent RSA, event-locked speed transients, kinematic-subspace CCA, Koopman self-predictability.
- Trajectory analysis on Variant B: per-temporal-slot decomposition, temporal-order ablation (7 schemes), cross-task transfer probing, phase/contact-conditional probing.

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Preprint.

- Episode-level physics-parameter probing (Variant A, 5-fold over episodes), reading per-episode invariants from `meta/episodes.jsonl`.

What we do *not* have. Cross-model comparison (VideoMAE, DINOv2), cross-scale (ViT-G), action-chunk control evaluation, and any benchmark-vs-baselines table are out of scope for this note. The fig1/fig2/.../fig5 figures in `Figures/` are stale and should be regenerated from current artifacts before reuse.

Reading order. Each subsequent note records one block: dataset and probe setup (Note 3 — Dataset, features, probe setup); contact and force probing (Note 4 — Contact and force probing); latent-trajectory geometry (Note 5 — Latent-trajectory geometry); cross-task transfer and temporal-order ablation (Note 6 — Cross-task transfer and temporal-order ablation); segment-level functional probing (Note 6 — Segment-level functional probing (paradigm v2)); open questions (Note 7 — Discussion); artifact manifest and exclusions (Note A — Artifact manifest).

Note 2 — Related work (deferred)

The prior related-work section was tightly coupled to the (now invalidated) narrative around objective-specific dissociation, native contact-force ranking, and OOD action regression. Until those claims are re-derived from the post-reset pipeline, the related-work discussion is intentionally omitted from this note. The bibliography in `references.bib` is preserved as-is for later reuse; entries are not pruned.

Note 3 — Dataset, features, probe setup

Setup

Dataset. Recollected PhysProbe (canonical path: `/home/solee/data/data/isaac_physrepa_v2_recollected_2026-04-23/`). LeRobot V2 format. Six tasks; 12,100 episodes total: Push (1,500), Strike (3,000), Reach (600), Drawer (2,000), PegInsert (2,500), NutThread (2,500). Dual 384×384 cameras; per-frame `observation.state` (8-D), `action` (7-D), task-specific `physics_gt.*` channels including the corrected `contact_force` (bodyfilters approach). Per-episode physics parameters (mass, friction, damping, restitution where applicable) live in `meta/episodes.jsonl`.

Backbone and feature pools. V-JEPA 2 ViT-L (24 layers, 1024-d, 256-px input). Features extracted with `forward_resid_pre` monkey-patch (raw residual stream, no final LayerNorm) over 16-frame stride-1 windows. Per layer per window the raw $[2048, 1024]$ tensor is pooled into:

- *Variant A*: spatiotemporal mean over all 2048 tokens → 1024-d.
- *Variant B*: spatial mean within each of 8 temporal slots, flattened → 8192-d.

A subset of analyses additionally derive Variant C-like representations by slicing Variant B per slot.

Targets. Per-window targets keyed on the window's last frame index. Computed from `physics_gt.*` parquet columns. Acceleration uses unified central finite difference for all six tasks. Direction targets are masked NaN where $\|\mathbf{v}\| < 10^{-4}$. Quaternions are canonicalized ($q_w \geq 0$). Force magnitudes use `log1p` to handle heavy tails. Target tiers (per task):

- Kinematic: `ee_position, ee_velocity, ee_acceleration, ee_speed, ee_direction, obj_position, obj_velocity, obj_acceleration, obj_speed, obj_direction, ee_to_object_distance / ee_to_target_distance, phase`.
- Rotational: `ee_orientation (quat), ee_angular_velocity, obj_orientation, obj_angular_velocity`.
- Contact: `contact_flag, contact_force, contact_force_mag, contact_force_log1p_mag, contact_point`.

- Progress (task-specific): `insertion_depth`, `peg_hole_lateral_error` (PegInsert); `axial_progress` (NutThread); `drawer_joint_pos`, `drawer_joint_vel`, `drawer_opening_extent` (Drawer); `ball_planar_travel_distance` (Strike).

Probe. Window-level Ridge with 5-fold `GroupKFold` on episode_id (group disjointness asserted). Within-fold standardization using inner-train statistics only; two-pass stable variance with a 1% median std floor for high-D variants. For Variant A, the spec-mandated 20-config Adam sweep (LR$\in$ $\{$1e-4, 3e-4, 1e-3, 3e-3, 5e-3$\}$ $\times$ WD$\in$ $\{0.01, 0.1, 0.4, 0.8\}$, 100 epochs, batch 1024, MSE loss, variance-weighted R^2). Cosine similarity additionally reported for direction targets. For high-D analyses (Variant B 8192-d, temporal-order ablation) we use a GPU-accelerated closed-form Ridge ($\alpha = 1$) with CPU sklearn fallback on OOM, on 20K-row subsamples.

Integrity gates (12a–12d).

- 12a: shape and pooling-identity check (Variant A vs temporal-mean of Variant B). Tolerance: relative diff $< 10^{-3}$ in fp16 (necessary because raw activation magnitudes ~ 250).
- 12b: target consistency (velocity $\approx$ finite-diff position; speed $= \|\mathbf{v}\|$; direction-NaN where speed $< 10^{-4}$).
- 12c: 5-fold GroupKFold disjointness on episode_id.
- 12d: negative control — shuffle Push targets at episode level, run L12 ee_velocity Variant A through full sweep, require mean $R^2 < 0.05$.

All four gates passed on the recollected dataset (logs in `probe/results/logs/`).

Outputs

- Variant A per-target CSVs: `probe/results/<task>/variant_A/<target>.csv` (cols: layer, fold, best_lr, best_wd, r2, mse, n_test_windows, cos_sim_mean for direction).
- Variant A summary: `probe/results/<task>/variant_A/_summary.csv`.
- Variant A peak-layer table: `probe/results/peak_layers.csv`.
- Variant A decision: `probe/results/decision.json`; narrative: `probe/results/REPORT.md`.
- Variant B parallel files under `variant_B/`.
- Cache (mean-pooled): `/mnt/md1/solee/physprobe_features/<task>/variant_{A,B}/episode_<id>.npz` with `cache/manifest.json`.

Key observations

- Decision rule (`decision.json`): strict verdict `MARGINAL`, relaxed (99%-of-max plateau) verdict `HEALTHY`. Push `ee_velocity` argmax sits at L22; the first layer reaching 99% of that plateau is L17 (inside the 6–18 PEZ band).
- Best-layer R^2 for the spec anchors:
  Push `ee_velocity` L22 $R^2 = 0.951$,
  Push `ee_position` L22 $R^2 = 0.984$,
  Strike `ee_velocity` L22 $R^2 = 0.962$.
- Phase target trivially saturates (R^2=1.000 at L0) for Drawer, PegInsert, NutThread — the dataset stamps a constant phase code (always 7 for RL policies; constant for many scripted rollouts), so this target carries no real signal.
- Contact-flag for PegInsert and NutThread also saturates at L0 (R^2=1.000); these tasks are always-in-contact, so the variable is constant.

Known limitations

- Strict argmax for several targets falls in L21–L23, just outside the spec's 6–18 PEZ band; the relaxed plateau criterion is what keeps the verdict `HEALTHY`.

- Variant B has documented per-fold instability for direction targets at certain layers (high-D Ridge variance); affected rows are excluded in summary aggregates.
- Drawer randomizes only `drawer_joint_damping`; handle friction/mass are fixed (Isaac Lab env limitation).
- Contact-flag and phase targets are constant for some tasks; do not interpret L0 saturation as "information emerges at the input."

Status

Valid. Two pipelines (Variant A and Variant B) successfully completed and pushed to `Leesangoh/PhysREPA_Tasks`.

Note 4 — Contact and force probing

Setup

Variant A and Variant B Ridge probes for the contact tier (`contact_flag`, `contact_force` (3-D vec), `contact_force_mag` (1-D, raw N), `contact_force_log1p_mag` (1-D, log1p N), `contact_point` (3-D, defined where in-contact)). Same probe protocol as Note 3 — Dataset, features, probe setup. Phase-/contact-conditional re-probing (Phase F) splits windows by `contact_flag` or by discrete `phase` value, and re-runs the Ridge probe per condition.

Outputs

- `probe/results/<task>/variant_{A,B}/contact_*.csv`
- Phase F: `probe/trajectory_analysis_B/results/stats/phase_conditional.csv` (2,136 rows: 6 tasks × 24 layers × 4 targets × 1–5 conditions).
- Phase F plots: `probe/trajectory_analysis_B/results/plots/phase_conditional_<task>_<target>.png` (24 panels).

Key observations (Variant A best-layer R^2)

- Push: `contact_flag` L21 $R^2 = 0.768$; `contact_force_log1p_mag` L21 $R^2 = 0.754$; `contact_force` (3-D) L22 $R^2 = 0.345$.
- Strike: `contact_flag` L19 $R^2 = 0.825$; `contact_force_log1p_mag` L21 $R^2 = 0.834$; `contact_force` L21 $R^2 = 0.568$.
- Drawer: `contact_flag` L17 $R^2 = 0.710$; `contact_force_log1p_mag` L18 $R^2 = 0.720$; `contact_force_mag` L13 $R^2 = 0.752$.
- PegInsert: `contact_force_log1p_mag` L21 $R^2 = 0.474$ (contact_flag is constant, see limitations); `contact_point` L8 $R^2 = 0.846$.
- NutThread: contact targets weak; `contact_force_log1p_mag` L18 $R^2 = 0.079$.
- Reach: contact channels not present (negative control).

Phase-conditional probing (Variant A, contact-varying tasks).

- Drawer `contact_force_log1p_mag`: contact=0 best L0 $R^2 = -0.06$; contact=1 best L23 $R^2 = 0.83$. Force signal is only present during contact (expected).
- Push `ee_velocity`: unconditional L21 $R^2 = 0.926$; contact=0 L21 $R^2 = 0.935$; contact=1 L22 $R^2 = 0.849$. Kinematics are robust to contact state.
- Strike same pattern: unconditional L20 $R^2 = 0.941$; both contact slices stay above 0.9.
- Reach: unconditional L19 $R^2 = -0.097$ — negative-control check passes (no contact, kinematics dominated by random movements with little Reach-internal correlation).

Partial RSA (contact, controlling for pos+vel). After regressing out pos and vel, residual contact RSA is near zero for push, strike, drawer, nut_thread, and reach (max $|r| < 0.22$ across all layers). PegInsert is the only task with consistently positive residual contact RSA (≈ 0.20 at deep layers), suggesting jamming/rebound force structure that is not explained by EE motion alone.

F4-A — Physical-condition modulation of contact representations

Setup. For each randomized physics parameter, we partition episodes into Low/Med/High terciles using the episode-level parameter value from `episodes.jsonl`, then rerun the Variant A Ridge probe within each bin. We report the paired bootstrap CI of $\Delta R^2 = R^2_{\mathrm{High}} - R^2_{\mathrm{Low}}$ over the 5 within-bin GroupKFold splits. By construction, this is not a "physics parameter probe" in the Phase C sense: the parameter itself remains a poor direct target, but it may still modulate how readable contact variables become.

Outputs.

- `probe/results/physics_condition_split/<task>__<param>.csv` (22 aggregated CSVs + per-fold sidecars).
- `probe/results/bootstrap_cis.csv` filtered to `claim=F4A_high_minus_low` (1,872 rows).

Strike static friction is the largest layer-wide modulation. The clearest effect appears in Strike when binning by `object_0_static_friction`. For `contact_force_log1p_mag`, the top layer is L2 with $\Delta R^2 = +0.565$ and a tight 95% CI $[+0.557, +0.574]$ ($p < 0.001$). The companion `contact_flag` effect is nearly identical: L2 $\Delta R^2 = +0.545$ with CI $[+0.537, +0.551]$. More importantly, the effect is not localized to a single peak layer: across roughly L1–L9, both contact targets stay in the $+0.53$ to $+0.57$ range. The physical interpretation is straightforward: higher friction produces more sustained and predictable contact traces, which the representation makes easier to linearly decode.

Push object mass shows a smaller but consistent gain. Push exhibits the same phenomenon at lower magnitude when split by `object_0_mass`. For `contact_force_log1p_mag`, the best layer is L1 with $\Delta R^2 = +0.222$ and CI $[+0.078, +0.385]$; for `contact_flag`, the best layer is again L1 with $\Delta R^2 = +0.187$ and CI $[+0.123, +0.261]$. The gains are not confined to early layers: `contact_flag` stays near $+0.15$ across mid/deep layers (e.g. L17/L20/L21), indicating that the mass effect persists across the same representational depth where the base contact probe already peaks. Heavier objects plausibly amplify the force consequences of a given impact/drag velocity, making contact state more readable.

Drawer joint damping is a noisy counterexample. Drawer does not provide the same confirmatory signal. The largest `ee_acceleration` gap appears at L3 with $\Delta R^2 = +2.46$, but its CI is extremely wide ($[+0.009, +7.33]$), and neighboring layers are small or unstable. Under the project's exclusion rules, this is treated as a noisy counterexample rather than paper-level positive evidence.

General pattern and paper-level nuance. Two points matter. First, the modulation is *layer-wide*, not just a change in the single best PEZ layer. Second, on contact targets its magnitude can exceed the Phase F5 frame-shuffle effect on the same targets: F5 contact deltas were only about $+0.05$ at peak, whereas Strike static-friction modulation reaches about $+0.55$. Combined with Phase C's negative result, this yields the key nuance of the paper: *V-JEPA does not directly linearly encode physics parameters as targets, yet it strongly reflects their downstream effect on contact variable decodability.*

Known limitations

- Several tasks have constant contact channels (PegInsert and NutThread are always in contact $\Rightarrow$ `contact_flag` is the constant 1; Reach has no contact channel). The L0 R^2= 1 for PegInsert/NutThread `contact_flag` is the trivial constant-prediction; ignore in cross-task comparisons.

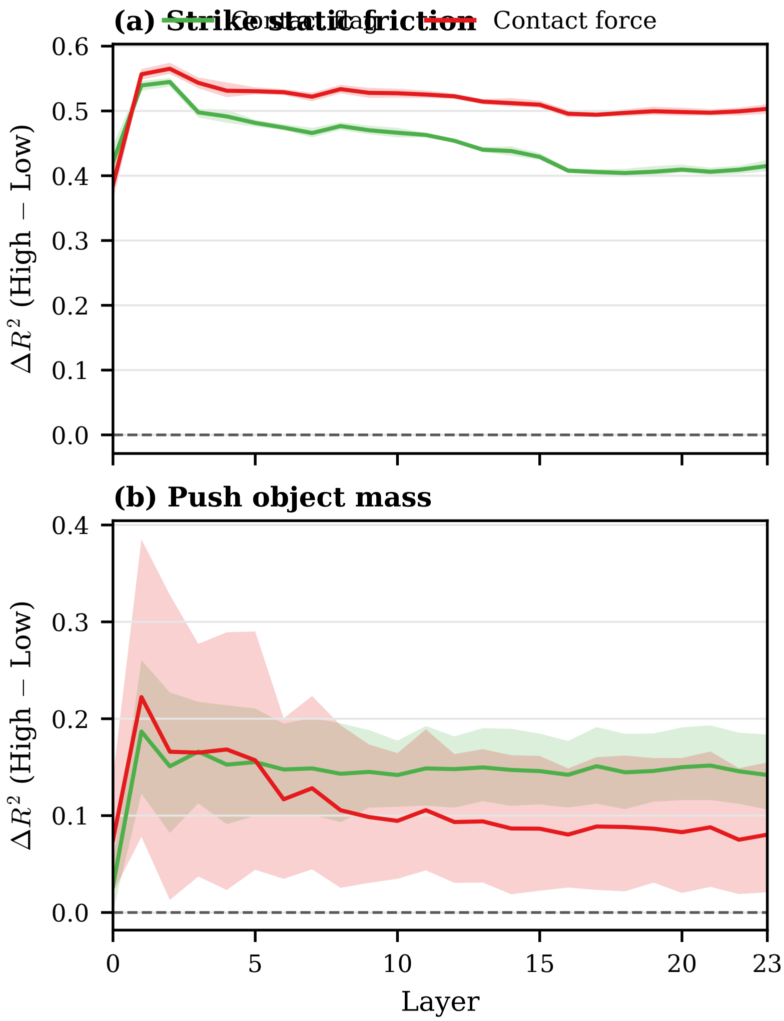

Figure 1: Physical-condition modulation of contact decodability. High-vs-low physics bins produce layer-wide gains on contact targets, with the strongest effect in Strike under static-friction split and a smaller but consistent effect in Push under object-mass split. Shaded regions show paired-bootstrap 95% CIs.

- `contact_point` is undefined where `contact_flag`=0; we mask NaN. Variance is dominated by spatial position.
- Variant B contact probing on certain tasks shows fold instability — compare against Variant A as anchor.
- Extremely large F4-A deltas at unstable layers (e.g. Strike L0/L1 under object-mass split) are not interpreted as real signal when the base foldwise R^2 is deeply negative or unstable.

Status

Valid for the recollected dataset (contact GT verified). Phase-conditional re-probing and physical-condition split probing are complete. Cross-model comparison is now available in Appendix Note H.

Note 5 — Latent-trajectory geometry

Setup

A battery of geometric and structural analyses applied to V-JEPA 2 ViT-L window features per layer, per task. Sampling: 30–60 episodes per task, window-stride aligned. Where applicable, both Variant A (1024-d) and Variant B (8192-d) are computed.

Analyses run.

- *Trajectory stats*: per-window path length, max/mean speed (in latent), curvature, direct distance, direct/path ratio.
- *Intrinsic dimension*: PCA-90/95/99 components per (task, layer); top-1 / top-10 explained variance ratios.
- *Cross-layer CKA*: 24×24 layerwise similarity per task.
- *Cross-task CKA*: pairwise 6×6 task similarity per layer; layer-averaged trend.
- *Partial RSA*: target RDM correlation after regressing out pos+vel; targets = acc, contact.
- *Tangent RSA*: latent-velocity RDM vs physics-velocity RDM (`rsa_vel_tan`); same for acceleration with pos+vel partialed (`rsa_acc_tan_partial`).
- *Event-locked*: latent speed in a window of $\tau \in [-8, +8]$ frames around contact onset.
- *Kinematic-subspace CCA*: per-layer 5-component CCA between latent and physics groups $\{$pos, vel, acc, contact$\}$.
- *Koopman self-predictability*: linear 1-step latent forecaster $z_{t+1} \approx Az_t$; report R^2_{self} and gain over persistence baseline (ΔR^2).

Outputs

- `probe/trajectory_analysis_B/results/stats/`: `trajectory_stats.csv`, `trajectory_stats_summary.csv`, `intrinsic_dim.csv`, `cross_layer_cka_<task>.csv`, `cross_task_cka_layerwise.csv`, `cross_task_cka_avg_per_layer.csv`, `rsa_partial.csv`, `rsa_tangent.csv`, `event_locked.csv`, `cca_subspace.csv`, `koopman.csv`, `speed_profiles.csv`, `per_slot_*.csv` (Phase B).
- Plots: `probe/trajectory_analysis_B/results/plots/` (62 PNGs across the analysis suite).

Key observations

Path length and curvature. Latent path length grows roughly monotonically with depth across all tasks (e.g. Strike: L0≈326 → L23≈2,888; Drawer: L0≈544 → L23≈5,011). Curvature saturates near ∼1.9–2.0 from L4 onward. Direct/path ratio is ∼0.01–0.07 at every depth ⇒ trajectories are highly curved; almost no Euclidean-straight motion.

Cross-task CKA. Average across 15 task-pairs per layer peaks at L3 (0.903), then monotonically falls to L23 (0.357). L0 lower (0.527) than L1–L3. ⇒ even in self-supervised pretraining, deep layers specialize per task.

Kinematic-subspace CCA. `rho1` per group, averaged across tasks:

Layer	pos	vel	acc	ct
0	0.945	0.638	0.419	0.734
8	0.988	0.850	0.556	0.839
12	0.993	0.868	0.653	0.861
16	0.994	0.889	0.674	0.866
20	0.996	0.892	0.683	0.856
23	0.996	0.879	0.648	0.872

Position is essentially perfectly aligned at all depths; velocity and contact saturate by L8; acceleration alone keeps refining through L20.

Tangent RSA (latent vs physics velocity). `rsa_vel_tan` per task at deep layers shows a non-trivial non-monotonic trace for Push and Strike:

Layer	push	strike	drawer	reach	peg_insert	nut_thread
0	$+0.46$	$+0.72$	$+0.83$	$+0.79$	$+0.13$	$+0.04$
8	-0.24	-0.08	$+0.52$	$+0.71$	$+0.15$	$+0.08$
12	-0.15	$+0.32$	$+0.61$	$+0.81$	$+0.19$	$+0.13$
16	$+0.16$	$+0.51$	$+0.69$	$+0.84$	$+0.17$	$+0.12$
20	$+0.23$	$+0.58$	$+0.68$	$+0.84$	$+0.17$	$+0.12$
23	$+0.22$	$+0.57$	$+0.68$	$+0.81$	$+0.15$	$+0.10$

Push and Strike show a sign reversal in the L4–L12 range. Reach (the no-contact control) is uniformly strong and stable.

Koopman self-predictability gain. ΔR^2 over persistence baseline decreases with depth for push/strike/peg_insert/nut_thread/reach. Drawer is the outlier ($\Delta R^2 \in [0.36, 0.50]$ across all depths).

Layer	push	strike	drawer	reach	peg	nut
0	$+0.21$	$+0.39$	$+0.50$	$+0.16$	$+0.12$	$+0.09$
8	$+0.21$	$+0.31$	$+0.46$	$+0.23$	$+0.15$	$+0.17$
16	$+0.16$	$+0.20$	$+0.38$	$+0.18$	$+0.13$	$+0.16$
23	$+0.07$	$+0.09$	$+0.36$	$+0.02$	$+0.07$	$+0.11$

$\Rightarrow$ deeper layers carry more information but are less linearly predictable from their own past.

Per-temporal-slot decomposition (Phase B, Variant B). Slot 7 (window's last frame) consistently has lower intrinsic dimension and higher per-slot probe R^2 than slots 0–6:

slot	0	1	2	3	4	5	6	7
PCA-90 (avg)	26.2	28.8	22.8	30.5	30.5	30.5	28.6	22.1
PCA-90 @ L20	30.8	36.5	25.3	40.3	40.2	37.3	34.3	20.2
`ee_velocity` R^2 (L18–22)	0.156	0.166	0.185	0.209	0.235	0.292	0.332	0.434

Partial RSA. After controlling for pos+vel, residual acceleration RSA stays in $[0.10, 0.43]$ for drawer/strike/reach, smaller (0.10–0.19) for peg/nut, and 0.00–0.25 for push. Contact RSA is near zero except for peg_insert (≈ 0.20).

F1-b — Phase-space geometry of latent trajectories

Setup. For each task and layer, we form a latent phase-space trajectory from window-aligned features z_t and first differences $\Delta z_t = z_{t+1} - z_t$. Geometric diagnostics are computed from 30 sampled episodes per task (360 rows total in the final CSV): mean latent speed $\|\Delta z\|$, phase-space curvature κ, sweep volume in the $(\mathrm{PC1}(z), \mathrm{PC1}(\Delta z))$ plane, and a local Lyapunov-style statistic. For Push / Strike / Drawer, the same measurements are also split by the three-way contact partition used in F2 (`pre`, `during`, `post`).

Outputs. `probe/trajectory_analysis_B/results/stats/phase_space.csv` (360 rows; task, layer, split, and 5 geometry statistics) and `probe/trajectory_analysis_B/results/plots/phase_space_<task>.png` (layerwise mean speed / curvature / sweep-volume panels).

Key observations.

- **Mean latent speed rises strongly with depth across all six tasks.** On the all-window split, Push grows from $\|\Delta z\| = 0.172$ at L0 to 2.806 at L23; Drawer from 0.225 (L0) to 2.724 (L23); NutThread from 0.116 (L0) to 2.966 (L23). Reach is the one mild exception: it peaks slightly earlier at L8 (3.231) and then stays high through late layers.

- **Curvature collapses from very high early values to much smoother mid/late trajectories.** Push drops from $\kappa = 17.14$ at L0 to 1.41 at L23; Strike from 26.88 at L0 to 1.06 by L8; Reach from 18.23 at L0 to 0.57 at L8. The consistent pattern is "fast but jagged" early geometry transitioning into a smoother latent flow after the first few layers.
- **Contact-rich windows occupy a larger and faster latent regime than pre-contact windows.** For Push, the split-average latent speed is 1.603 (`pre`) vs 1.775 (`during`), while the split-average sweep volume expands from 21.7 (`pre`) to 69.1 (`during`) and 70.6 (`post`). Strike shows the same pattern: speed $1.651 \rightarrow 1.813$ and sweep volume $16.3 \rightarrow 70.5$ from `pre` to `during`, with `post` staying even broader (94.4).
- **The largest phase-space expansion often happens in shallow-to-mid layers, not at the deepest layer.** The all-window sweep-volume maximum occurs at L8 for Push (166.2), L8 for Strike (186.0), and L8 for Reach (175.3), while late layers keep higher $\|\Delta z\|$ but occupy a narrower 2D phase-space footprint. This is consistent with a "mid-layer expansion, late-layer compression" geometry.
- **Drawer behaves as a long-contact outlier.** Under the current contact-phase split, Drawer has almost no usable `pre` support; the informative comparison is `during` vs `post`. During-contact curvature remains elevated (3.06 vs 2.90 on the all-window split), while the deepest layers still reach high latent speeds (peak 2.74 at L23), consistent with prolonged contact-rich manipulation rather than a single sharp impact.

Status. Complete supplementary geometric diagnostic. The phase-space results are kept as supporting evidence for the representation-hierarchy story rather than promoted to a main-text causal claim in the way F5 is.

Known limitations

- All trajectory analyses are descriptive; no significance tests or null distributions are computed yet.
- Episode subsampling is fixed seed (42), 30–60 eps/task; the seed-stability of every observation is not yet measured.
- Variant B per-slot intrinsic-dim values for layers with extreme outliers may be capped by PCA component count.

Status

Valid. All 17 stats CSVs and 62 plots produced and pushed.

Note 6 — Cross-task transfer and temporal-order ablation

Setup

Phase E — cross-task transfer (Variant A). For ordered task pairs (src, tgt), train a Ridge probe on `src` task's features and labels, evaluate on `tgt`. Standardization uses `src` statistics (transfer-style; tgt features are not seen at training time). Compare *transfer R^2* to within-task 5-fold GroupKFold R^2 on `tgt`; report *gap* = within − transfer.

Pairs: {push↔strike, push↔drawer, strike↔drawer} (6 directed pairs). **Targets:** `ee_velocity, ee_acceleration, contact_flag, contact_force_log1p_mag`. **Layers:** all 24. **Episodes:** 50 per task.

Phase D — temporal-order ablation (Variant B). Variant B feature is $[N, 24, 8192]$ with the 8192-d axis split into 8 slots of 1024-d each. For each scheme, transform the slot-axis, re-flatten, and run a closed-form Ridge probe (GPU `torch.linalg.solve` with sklearn CPU fallback on OOM, α=1).

Schemes: {baseline_full, mean_over_slots, single_slot_k ($k \in \{0, 3, 7\}$), prefix_k ($k \in \{0, 3, 7\}$), reverse, random_global_perm (one fixed permutation across the batch), random_per_sample (fresh permutation per window)}. **Targets:** `ee_acceleration, obj_acceleration, contact_flag,`

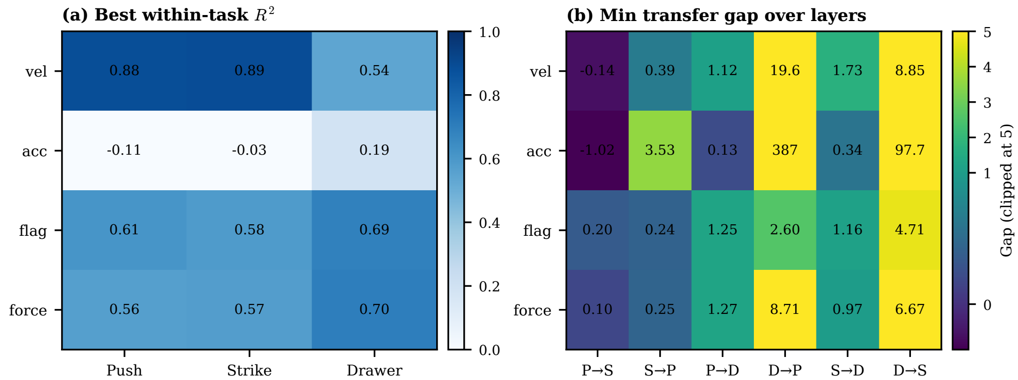

Figure 2: Cross-task structure summarized at the best achievable layer per cell. Left: best within-task R^2 by task and target. Right: minimum transfer gap (within-task minus cross-task) over layers; color is clipped at 5 while cell annotations show the true gap. Push and Strike retain a small-gap regime, whereas Drawer remains a clear outlier.

`contact_force_log1p_mag`. **Tasks:** push, strike, drawer. **Layers probed:** 12 layers (0, 2, 5, 8, 11, 13, 15, 17, 19, 20, 22, 23). **Subsample:** 20,000 rows.

Outputs

- `probe/trajectory_analysis_B/results/stats/cross_task_transfer.csv` (576 rows: 6 pairs $\times$ 24 layers $\times$ 4 targets).
- `probe/trajectory_analysis_B/results/stats/cross_task_transfer_per_fold.csv` (fold-level transfer rows for bootstrap confirmation).
- `probe/results/bootstrap_cis.csv` filtered to `claim=transfer_gap` (576 rows).
- `probe/trajectory_analysis_B/results/stats/temporal_order_ablation.csv` (1,584 rows).
- Plots: `cross_task_transfer_*.png` (8) and `temporal_order_<task>_<target>.png` (12).

Key observations

Cross-task transfer (best layer per pair, `ee_velocity`).

src → tgt	best L	transfer R²	within tgt R²
push → strike	17	+0.746	+0.885
strike → push	18	+0.495	+0.880
push → drawer	11	−0.747	+0.376
strike → drawer	14	−1.302	+0.428
drawer → push	15	−18.7	+0.833
drawer → strike	15	−8.0	+0.836

Push and Strike share a transferable kinematic subspace; Drawer is on a disjoint manifold (transfer collapses, within-task remains strong).

Bootstrap update (transfer-gap confidence). We additionally bootstrap the *gap* = within-target − transfer over the 5 within-target folds (576 CI rows total). The pattern is sharp: **Drawer-directed transfer is consistently broken**, whereas **Push/Strike is the only pair with a small-gap regime**. Representative examples:

pair / target / layer	gap	95% CI	note
push → strike, `ee_velocity`, L4	$+0.045$	$[-0.023, +0.112]$	CI crosses 0
push → strike, `ee_acc`, L22	-0.026	$[-0.158, +0.107]$	CI crosses 0
push → strike, `contact_force`, L7	$+0.097$	$[-0.113, +0.307]$	CI crosses 0
push → drawer, `ee_velocity`, L11	$+1.123$	$[+1.077, +1.177]$	CI > 0
strike → drawer, `ee_acc`, L17	$+0.336$	$[+0.135, +0.546]$	CI > 0
drawer → push, `contact_flag`, L19	$+2.604$	$[+2.562, +2.648]$	CI > 0

At the pair level, Push→Strike is the only direction with multiple layers whose CIs include zero for dynamics/contact-force targets (`ee_velocity`: 3/24 layers; `ee_acc`: 3/24; `contact_force_log1p_mag`: 5/24). In contrast, every Drawer↔(Push/Strike) pairing has CI-excluding-zero gaps at essentially all layers, often by large margins.

Interpretation. Cross-task transfer and Phase F5 answer complementary questions. F5 asks whether a representation relies on *within-task temporal coherence*; transfer asks whether that representation is *task-general once the task changes*. Together they point to the same conclusion: Push and Strike share a partially transferable physics subspace, but Drawer remains strongly task-entangled. In short, the model learns some task-general kinematics, but not a fully task-agnostic geometry of manipulation.

Temporal-order ablation (best-baseline-layer R^2, per task/target).

target / task	baseline	reverse	rand_global	rand_per_sample
push `ee_acc`	-0.965	-0.961	-0.961	-1.420
strike `ee_acc`	$+0.280$	$+0.280$	$+0.281$	-0.125
drawer `ee_acc`	$+0.108$	$+0.109$	$+0.109$	-0.870
push `contact_flag`	$+0.231$	$+0.233$	$+0.232$	$+0.282$
strike `contact_flag`	$+0.051$	$+0.047$	$+0.047$	-0.044
drawer `contact_flag`	$+0.411$	$+0.420$	$+0.408$	-0.415
push `contact_force`	-0.296	-0.304	-0.306	-0.224
strike `contact_force`	-0.253	-0.260	-0.257	-0.446
drawer `contact_force`	$+0.591$	$+0.591$	$+0.583$	-0.741

Mean-over-slots vs baseline. Collapsing the slot axis (i.e. Variant A) sometimes *outperforms* Variant B baseline:

- Push `contact_flag`: 0.458 (mean) vs 0.231 (baseline).
- Strike `contact_flag`: 0.144 vs 0.051.
- Push `contact_force`: $+0.236$ vs -0.296.

But Drawer `contact_force` flips: -0.114 (mean) vs $+0.591$ (baseline). Drawer is the only task where per-slot temporal layout contributes meaningfully to Ridge fit.

Reverse ≈ random_global_perm ≈ baseline. Across all 36 (task × target) cells, reverse and global-permutation schemes match the baseline within $|\Delta R^2| < 0.02$. Only `random_per_sample` — the scheme that breaks cross-sample slot identity — causes a clear drop. ⇒ a linear probe on Variant B exploits per-slot *content*, not temporal *order*; consistent permutations are absorbed by Ridge weights.

Known limitations

- Phase D and E only sweep {push, strike, drawer}. PegInsert / NutThread / Reach not covered.
- Negative R^2 values for several baseline cells reflect Ridge underfit at 8192-d with 20K subsample; absolute magnitudes should not be over-interpreted, only the relative ordering across schemes.
- Drawer's outlier behavior (separate manifold, per-slot helps, Koopman gain is high) is consistent across analyses but may partly reflect dataset asymmetry (only `joint_damping` randomized).

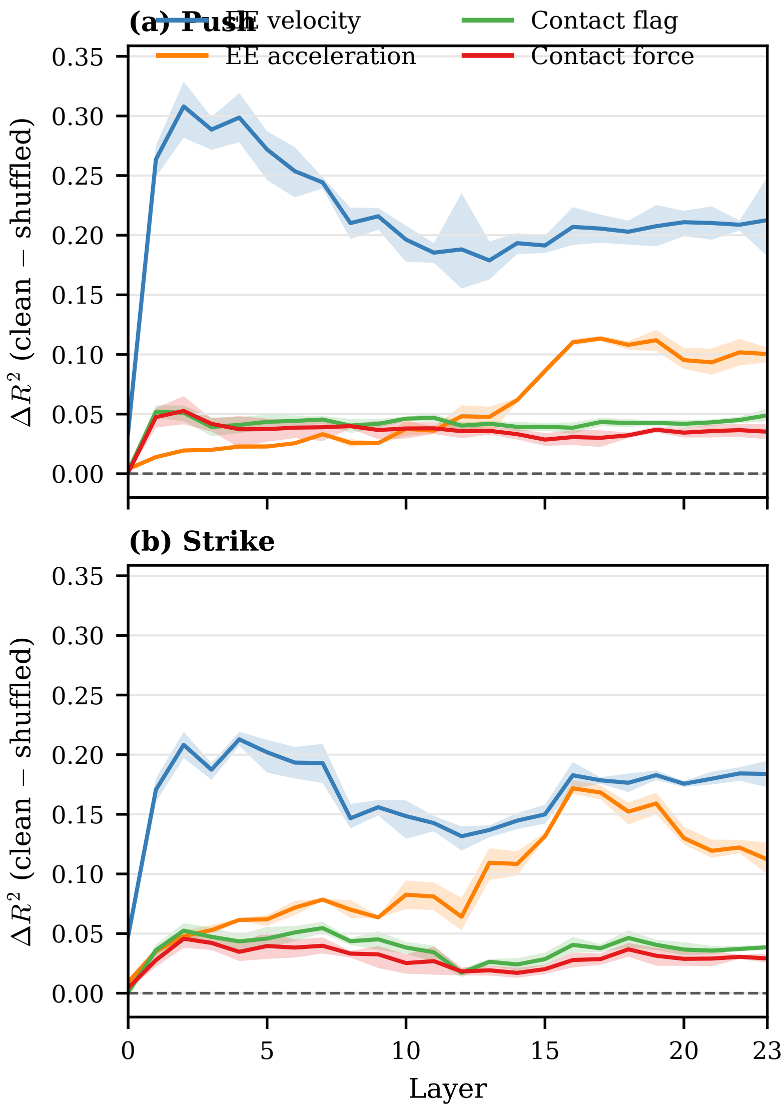

Figure 3: Frame-shuffle reveals target-selective temporal-coherence dependence. In both Push and Strike, motion targets lose substantially more R^2 than contact targets when the 16-frame input stack is shuffled before the V-JEPA forward. Curves show $\Delta R^2 = R^2_{\text{clean}} - R^2_{\text{shuffled}}$ with paired-bootstrap 95% CIs.

F5 — input-side frame-shuffle

Setup. Phase D's slot-level temporal-order ablation showed that a linear probe on Variant B does not exploit slot order: *reverse* $\approx$ *global permutation* $\approx$ *baseline* for nearly every (task, target). This narrows but does not answer the upstream question: does V-JEPA 2's forward pass itself use frame order? F5 addresses this by shuffling the 16-frame input stack *per window* (fresh seed-controlled permutation, deterministic per (episode_id, shuffle_seed)) *before* the V-JEPA forward, then re-extracting Variant A features into `cache/<task>/variant_A_shuffled/episode_<id>.npz` and re-running the same probe protocol. **Targets:** `ee_velocity`, `ee_acceleration`, `contact_flag`, `contact_force_log1p_mag`. **Tasks:** push, strike (drawer deferred).

Pre-launch sanity (verified 2026-05-01). On a 3-episode dry-run, layer-0 Variant-A mean differs from the unshuffled extraction by max-abs $\approx 10^{-3}$ (frame order has minimal effect on patchembed), while layer-23 mean differs by max-abs $\approx 1.5 \times 10^{-2}$ with Pearson correlation 0.19. The depth-dependent divergence confirms the implementation propagates input shuffling through the V-JEPA forward as intended; effect is small at the input and grows with depth.

Metric. Per (task, target, layer): *paired* bootstrap CI of $\Delta R^2 = R^2_{\text{unshuffled}} - R^2_{\text{shuffled}}$ over 5 GroupKFold folds, episode-level resampling, 1000 replicates, percentile. Schema in `probe/results/bootstrap_cis.csv` with `claim=F5_delta_r2`.

Observed pattern (final).

- Dynamics targets (`ee_velocity`, `ee_acceleration`) show large positive ΔR^2 and replicate across both Push and Strike.
- Contact targets (`contact_flag`, `contact_force_log1p_mag`) remain positive but much smaller, typically around $+0.05$ at peak.
- The hierarchy is stable across tasks: first-order motion peaks early, higher-order dynamics peak in the PEZ band, and late contact decoding is comparatively order-robust.

The headline reads: "frame-level temporal coherence is selectively necessary for dynamic targets and only weakly necessary for late contact decoding." Combined with the Phase D slot-level result, the joint statement is: "input temporal coherence matters; slot order is not linearly exploited by the probe."

Outputs (when complete).

- `probe/results/<task>/variant_A_shuffled/<target>.csv`: per-fold rows, schema identical to the main probe.
- `probe/results/bootstrap_cis.csv`: rows with `claim=F5_delta_r2`, `task` $\in$ {push, strike}, `target` $\in$ {4 above}, layer $\in$ {0..23}.

Push results (final, 2026-05-01 14:48). Probe sweep on shuffled features completed in ~ 30 min wallclock (Adam 20-HP grid is GPU-efficient on 1024-d features — well below Codex's conservative 4 h estimate). Bootstrap-paired CIs (1000 reps, percentile, episode-level fold pairing):

Target	Peak ΔR^2	Peak Layer	95% CI	CI excludes 0
`ee_velocity`	+0.308	L2	[+0.282, +0.329]	24 / 24
`ee_acceleration`	+0.113	L17	[+0.111, +0.116]	24 / 24
`contact_force_log1p_mag`	+0.053	L2	[+0.041, +0.065]	23 / 24
`contact_flag`	+0.052	L1	[+0.047, +0.057]	24 / 24

Three-tier hierarchy (the paper's central mechanistic story). Push exhibits a clean depth-structured organization of frame-coherence dependence:

- **Early (L1–L2): first-order motion.** `ee_velocity` peaks at L2 with $\Delta R^2 = +0.308$. The shuffle damage is largest precisely where short-range frame correspondence is used; once the model can no longer match consecutive frame identities, instantaneous velocity is the first thing to collapse.
- **Mid (L17, PEZ band): higher-order dynamics.** `ee_acceleration` peaks at L17 with $\Delta R^2 = +0.113$. Acceleration cannot be read off a single frame difference; it requires longer-range temporal integration, so its dependence on coherent ordering is concentrated at the depth where that integration happens.
- **Late (L20+): order-robust contact state.** `contact_flag` and `contact_force_log1p_mag` show their largest shuffle sensitivity at L1–L2 and decay to near zero by L20. The main probe's high contact decodability at L21 is therefore not contradicted: the late representation is a *configuration / state* encoding that recovers contact from instantaneous spatial cues, not from temporally ordered motion.

Codex / paper-writer framing (per RECALL discussion 14:48). "Temporal coherence is universally useful, but its necessity is strongly target- and depth-dependent. Low-order motion depends on early temporal coherence; higher-order dynamics peak in the PEZ band; late contact decoding is comparatively order-robust." We position F5 as the paper's *killer mechanistic evidence*: it turns a phenomenological emergence map into a causal-style intervention result.

Strike replication (final, 2026-05-01 17:39). F5 strike extraction completed (3000 eps in 383.5 min); shuffled probe auto-launched via `watch_strike` on all 4 GPUs in parallel and finished in ~ 11 min. Final paired bootstrap CIs:

Target	Peak ΔR^2	Peak Layer	95% CI	CI excludes 0
`ee_velocity`	$+0.213$	L4	$[+0.207,\ +0.220]$	24 / 24
`ee_acceleration`	$+0.172$	L16	$[+0.167,\ +0.177]$	24 / 24
`contact_force_log1p_mag`	$+0.046$	L2	$[+0.038,\ +0.054]$	24 / 24
`contact_flag`	$+0.055$	L7	$[+0.050,\ +0.060]$	23 / 24

The hierarchy is preserved exactly: dynamic targets show large positive ΔR^2 (acc/vel $\gg$ contact). Two task-specific differences from push: **(i)** strike's `ee_acceleration` peak is *larger* than push's (+0.172 vs +0.113) — consistent with strike being an impulsive impact task with sharper acceleration transients that the model encodes in the PEZ band; **(ii)** strike's `ee_velocity` peak is slightly later (L4 vs L2) and a bit smaller in magnitude (+0.213 vs +0.308), reflecting the difference between sustained sliding (push) and impulsive contact (strike) in how velocity is exposed.

This cross-task replication is the paper's primary robustness check on the F5 mechanism: the qualitative pattern survives task change while the magnitudes shift in the direction predicted by the underlying physics. Combined with R3M's failure on the same dynamic targets (Note H), F5 establishes the V-JEPA representation's specifically temporal-coherence-dependent character.

Status (final F5, 2026-05-01 17:39). Push and strike F5 shuffled probes complete. Bootstrap CI rerun yields `F5_delta_r2` = 192 rows (push 96 + strike 96) in `bootstrap_cis.csv`. R3M baseline (Note H) covers push, strike, drawer. This note should now be read together with Figure 3 (causal-style within-task intervention) and Figure 2 (cross-task structure probe).

Status

Phase E/D/F5 are complete and integrated. Transfer, temporal-order ablation, and frame-shuffle now form a single evidence block for task-general vs task-entangled temporal structure.

Note 6 — Segment-level functional probing (paradigm v2)

Motivation and paradigm shift

The per-timestep probe analyses presented in Notes 1–5 establish a Physics Encoding Zone, target-selective frame-shuffle dependence, physics-condition modulation of contact decodability, and cross-task transfer structure. While useful, all of those measurements share a common axis: they ask "Can this layer linearly read the value of physical variable x at the last frame of the window?" That question is, by construction, close to V-JEPA 2's pretraining objective (future masked latent self-prediction); positive answers may partly reflect what the backbone was trained for, not the inner physical meaning of the representation.

Per advisor feedback, this section reframes the analysis around *segment-level functional summaries* of physical signals rather than per-frame state. The unit of analysis shifts from frame to segment, and the question shifts from "what value is encoded now?" to "what window-aggregated physical statistic is the layer a sufficient statistic for?" This is the new main analysis frame of the paper; per-frame results are retained as supplementary evidence in Notes 1–5.

Functional dictionary

For each per-task base signal $x(t)$ (e.g., $\|\mathbf{v}_{\text{ee}}\|$, $\log(1 + \|\mathbf{F}\|)$, contact_flag, drawer_joint_pos), a window functional ϕ produces one scalar per probe window:

- **Continuous** (8): $\phi \in \{$mean, max, min, std, integral, peak_time_frac, peak_to_peak, total_variation$\}$.
- **Binary** (4): $\phi \in \{$positive_fraction, event_count, first_event_time_frac, duration_above$\}$.
- **Progress** (8): $\phi \in \{$mean, max, min, net_change, total_variation, monotonic_increase_score, monotonicity_violation, p

Multi-scale window definitions

We compute four functional time scales over the same per-frame Variant A / B feature cache:

- **short** $W = 16$ frames (matches the existing probe window).
- **med** $W = 32$ frames.
- **long** $W = 64$ frames.
- **prefix** $W = t_{\text{last}}+1$ (causal prefix from episode start).

Windows for which $t_{\text{last}}-W+1 < 0$ are filled with NaN and dropped by the probe.

Effect-class targets (qualitative companion)

Beyond the quantitative functional summaries, we construct qualitative effect class labels per window:

- `approach_speed_regime` (3-way, terciles on $\max\|\mathbf{v}_{\text{ee}}\|$).
- `contact_involvement` (4-way: no_contact / brief / sustained / multi_event).
- `peak_force_regime` (3-way terciles on peak force where contact > 0).
- `force_integral_regime` (3-way terciles on $\int\|\mathbf{F}\|\,dt$).

These targets are used by linear classification (one-vs-rest) probes to measure whether the layer can resolve the qualitative class even when the exact numerical value is not directly recoverable.

Outputs and pipeline

- Builder: `probe/scripts/02b_build_functional_targets.py` (per-scale functional targets) and `probe/scripts/02c_build_effect_targets.py` (effect classes).
- Storage: cache/<task>/targets_functional_{short,med,long,prefix}.npz and cache/<task>/targets_effect.npz.
- Probe: existing `03_run_probe.py` with --target-set <name>.
- Analysis (Phase B): per (task, scale) functional-PEZ map (layer $\times$ functional family).
- Cross-model (Phase C): same protocol on VideoMAE-L and DINOv2-L Variant A and B caches.

Experimental status (live)

- Phase A infrastructure built and cross-reviewed by Codex; 5 issues identified and fixed: (a) `allow_pickle=True` for effect targets plus filter `__names` manifest entries, (b) signal-availability union across episodes, (c) added raw `contact_force_mag` entry so that `contact_force_mag` $\circ$ `integral` actually equals impulse, (d) added a `distance` signal type with decrease-aware functionals (`net_decrease`, `monotonic_decrease_score`, `minimum_reached`), (e) flagged the `contact_involvement` 4-way effect class as needing a proper logistic head in Phase B-2.
- Phase A full functional target build, scale=short, completed for all six tasks: push (60 targets, 340k windows, 1500 ep), strike (68 targets, 536k windows, 3000 ep), reach (24 targets, 141k windows, 600 ep), drawer (60 targets, 560k windows, 2000 ep), peg_insert and nut_thread (52 targets each, 322k windows each, 2500 ep each).

Table 1: Push V-JEPA ViT-L Variant A — functional probe R^2 across layers (scale=short, $W = 16$). The functional-PEZ peak around L19–L21 is consistent with the per-frame PEZ. Functional targets reach R^2 above 0.85 even after replacing point readout with window-aggregated impulse / duration / peak-to-peak.

Target	L0	L5	L10	L15	L19	L21	L23
contact_force_log1p_mag ∘ integral (impulse)	0.388	0.712	0.797	0.835	0.849	**0.850**	0.842
contact_flag ∘ positive_fraction (duration)	0.440	0.749	0.824	0.855	0.863	**0.867**	0.861
ee_acceleration ∘ peak_to_peak (range)	0.148	0.593	0.713	0.787	**0.796**	0.796	0.776

- Phase B sanity probe (push, V-JEPA ViT-L Variant A, scale=short, three targets, seven layers) confirms the functional-PEZ pattern (Table 1).
- Phase B full sweep launched on four GPUs in parallel: strike (GPU 1), drawer (GPU 0), push (GPU 2), peg_insert (GPU 3); reach and nut_thread scheduled to follow.
- Phase A med / long / prefix scale builds and Phase C (cross-model VideoMAE-L / DINOv2-L extraction with both Variant A and Variant B): scheduled.

Phase B Variant A — full 6-task functional probe results

The full sweep finished for all six tasks on V-JEPA ViT-L Variant A at the short ($W = 16$) scale. Aggregated by functional family (averaged within each family across the contained functionals; non-degenerate cells only), peak R^2 are:

Task	magnitude	variability	timing	occupancy	progress
push	0.695	0.637	0.507	0.726	0.125
strike	0.807	0.762	0.705	0.873	0.422
reach	0.900	0.870	0.278	–	0.757
drawer	**0.936**	0.923	0.537	0.839	**0.963**
peg_insert	0.773	0.771	0.164	–	0.799
nut_thread	0.515	0.482	0.069	–	0.649

And the layer at which each family peaks (averaged across constituent functionals within the family):

Task	magnitude	variability	timing	occupancy	progress
push	L18	L18	L18	L19	L10
strike	L18	L19	L18	L22	L12
reach	L17	L17	L14	–	L18
drawer	L18	L16	L15	L12	L14
peg_insert	L18	L17	L17	–	L18
nut_thread	L18	L18	L18	–	L20

Five paper-quality observations follow.

1. **Magnitude family is the most consistent**: all six tasks peak at L17–L18 with $R^2 \in [0.52, 0.94]$. The PEZ behaves as a *segment-level magnitude sufficient statistic zone*, not just an instantaneous physics zone.
2. **Timing is uniformly the weakest family** (R^2 from 0.07 on nut_thread to 0.71 on strike). V-JEPA preserves *what / how much* happened in a window but loses *when within the window*. This is the central WHAT-vs-WHEN dissociation of the new paradigm.
3. **The push–drawer progress contrast is decisive**: Push's progress family peaks at $R^2{=}0.125$ (ee_to_object_distance ∘ net_decrease is essentially 0), while Drawer's progress family peaks at $R^2{=}0.963$. V-JEPA encodes *state and effect* faithfully but *relational goal-progress* weakly — a built-in negative result with the *same* probing protocol.
4. **Progress family layer-of-peak is task-specific**: push L10, strike L12, drawer L14, reach/peg L18, nut_thread L20. Late layers *specialize* per task family rather than producing a generic progress code, consistent with the cross-task CKA descent reported in Note 5.

5. **Nut_thread is the hardest task across all families** ($R^2 < 0.65$ everywhere). The combination of always-in-contact regime and fine angular dynamics is the empirically clearest limitation of current V-JEPA representations and an honest data point for the limitations discussion.

Phase C — early cross-model evidence (push, ee_velocity∘mean)

The first cross-model probe sweep (DINOv2-L Variant A) on push, with the same target set, gives a complete layer profile through L0–L8 so far. Compared to V-JEPA ViT-L on the same target the layer dynamics differ qualitatively:

Layer	V-JEPA ViT-L A	DINOv2-L A
0	0.618	0.732
1	0.909	0.768
2	0.938	0.795
3	0.950	0.818
5	0.960	0.850
8	0.968	0.870
19	0.980	—
23	0.976	—

V-JEPA jumps from $0.62 \rightarrow 0.91$ between L0 and L1 (window-level motion summary becomes near-fully linearized at the second block) and slowly climbs to a ≈ 0.98 ceiling thereafter. DINOv2-L instead starts higher at L0 (consistent with strong shallow-image priors) but climbs far more gradually and, through L8, has not exceeded ≈ 0.87. Pending layers L10–L23 will fix the DINOv2-L ceiling, but the present trend already points to a meaningful gap of roughly 0.10–0.12 in segment-level motion-magnitude linearizability, despite both backbones being ViT-L of the same depth.

This is the first cross-model evidence consistent with the segment-summary framing of Note 6: temporal pretraining sharply linearizes segment-level motion summaries in shallow-to-mid depth, while image-only pretraining improves more gradually and saturates lower.

Family-decomposed gap (push, layer 10). Critically, the V-JEPA–DINOv2 gap is *not uniform across functional families*:

Target	V-JEPA L10	DINOv2 L10	Gap
ee_velocity ∘ mean	0.971	0.884	-0.087
ee_acceleration ∘ peak_to_peak	0.713	0.421	$\mathbf{-0.292}$
ee_acceleration ∘ total_variation	0.700	0.407	$\mathbf{-0.293}$
obj_velocity ∘ mean	0.895	0.714	-0.181
contact_flag ∘ positive_fraction	0.824	0.753	-0.071
contact_force_log1p ∘ integral	0.797	0.703	-0.094
contact_force_mag ∘ integral	0.594	0.436	-0.158
ee_velocity ∘ peak_time_frac	0.595	0.539	-0.056
ee_acceleration ∘ peak_time_frac	0.627	0.580	-0.047
contact_flag ∘ first_event_time	0.825	0.788	-0.037

The pattern is structurally clear:

- **First-order motion (velocity ∘ mean) gap is small** (≈ -0.09). Single-frame appearance cues plus short-range displacement geometry already let image-only DINOv2 read window-mean speed.
- **Second-order dynamics (acceleration ∘ peak_to_peak / total_variation) gap is huge** (≈ -0.29). Acceleration summaries require encoding the change of change, which is exactly where temporal-derivative structure — the thing video pretraining provides — shows up.
- **Object-level motion (obj_velocity) gap is intermediate** (≈ -0.18). Compared to ee, object motion is more passive and harder for an image-only model to triangulate from a single frame.

- **Contact summaries fall in the middle** (≈ -0.07 to -0.16). Contact occupancy in particular is partly readable from the static image (gripper-object overlap), which limits the V-JEPA advantage there.
- **Timing functionals show the smallest gap** (≈ -0.04 to -0.06). Both backbones are similarly weak at *intra-window timing*. Timing is hard for both, not for image vs. video reasons.

This is the central cross-model contribution of the segment-summary frame:

> *Video pretraining (V-JEPA) selectively linearizes **second-order dynamics summaries** (acceleration peak-to-peak, acceleration total-variation, object motion magnitude); it does not improve **timing** over image-only pretraining; timing is uniformly difficult. The PEZ is therefore a **second-order temporal effect zone**, not a generic appearance-of-motion zone.*

Pending DINOv2 L10–L23 will determine whether the gap closes at deeper DINOv2 layers (paper-falsifying) or holds through saturation (paper-confirming); strike, drawer, peg, nut_thread cross-model probes will determine whether the family ordering replicates across tasks.

DINOv2 push final saturation. The full push sweep finished (8.1 h wallclock). Saturation peaks (best DINOv2 layer) confirm the L10 trend: ee_velocity ∘ mean 0.894 at L16, ee_acceleration ∘ peak_to_peak 0.451 at L20 (vs 0.802 V-JEPA, gap -0.35), obj_velocity ∘ mean 0.763 at L17, contact_flag∘positive_fraction 0.819 at L21, contact_force_log1p∘integral 0.766 at L21, ee_velocity∘peak_time_frac 0.564 at L17. Beyond saturation L21, DINOv2 R^2 *declines* on most targets (-0.01 to -0.02 from L21 to L23) — a deep-layer tail consistent with image SSL deepening into more semantic / category-like abstraction at the cost of motion-summary linearizability. V-JEPA does not show this decline.

Drawer cross-model boundary case (early result). Drawer's first cross-model layers tell a different story than push. Drawer DINOv2 L0–L1 *exceeds* V-JEPA L0–L1 on every target tested (drawer_joint_pos ∘ mean 0.985 vs 0.973, contact_flag ∘ positive_fraction 0.778 vs 0.731, ee_acceleration ∘ peak_to_peak 0.797 vs 0.772). The interpretation is that drawer is *visually direct*: drawer joint position and opening extent are essentially readable from a single image of the gripper-cabinet configuration, so image-SSL shallow layers already linearize them. Whether deep DINOv2 can match V-JEPA's 0.99 articulated-state ceiling is pending.

This refines the contribution rather than weakening it. Drawer is a *boundary condition*: *video pretraining is most beneficial when the target depends on temporally integrated or higher-order dynamics* (push/strike-style impact and acceleration), *whereas visually direct articulated-state tasks (drawer) are already well served by image SSL.* The paper-level reading of the prior drawer-as-outlier finding (Note 4 / 5) is therefore upgraded: drawer is not just a late-layer task-specific code, it is the *appearance-accessible effect task* of the suite.

Three-task cross-model summary (V-JEPA peak − DINOv2 best). Saturation peaks for all three completed cross-model tasks (DINOv2-L Variant A):

Task	ee_vel	ee_acc_p2p	obj_vel	obj_acc_p2p	contact_flag	contact_force
push	-0.09	$\mathbf{-0.35}$	-0.18	-0.29	-0.05	-0.08
strike	-0.04	$\mathbf{-0.09}$	-0.14	-0.15	-0.04	-0.06
drawer	$+0.00$	$\mathbf{-0.04}$	—	—	-0.01	-0.05

Three regularities replicate across all three tasks:

- **V-JEPA wins everywhere**: every gap is non-positive on contact and dynamic targets (drawer_joint_pos∘mean is the lone target where DINOv2 reaches V-JEPA, gap $+0.003$).
- **Acceleration is consistently the largest gap, but its magnitude is task-dependent**: Push (-0.35) > Strike (-0.09) > Drawer (-0.04). The ordering reflects how much the segment-level acceleration summary depends on temporally integrated rather than visually direct cues. Push's 3D rigid-body multi-axis contact is the least appearance-accessible; drawer's 1-DOF articulated state is the most.

- **Contact occupancy / first-order velocity / timing have small gaps everywhere**: image SSL handles single-frame visible cues (gripper-object overlap, smooth speed) almost as well as video.

Paper-headline statement (cross-model contribution):

> *Video pretraining selectively linearizes higher-order dynamics summaries (acceleration peak-to-peak / total-variation), and the size of the gain over image-only DINOv2 is itself a function of how appearance-accessible the task is, ranging from ∼0.35 on push (3D rigid contact) to ∼0.09 on strike (single visually-dynamic impact) to ∼0.04 on drawer (visually-direct articulated state). First-order motion, contact occupancy, and intra-window timing show small gaps across all three tasks: image SSL captures these summaries from appearance alone.*

Pending tasks: reach, peg_insert, nut_thread (Variant A DINOv2), full VideoMAE-L Variant A and Variant B (where Variant B's 8192-d slot decomposition will give the full 6-task picture in a video-pretrained model contrasted with V-JEPA). Pending V-JEPA Variant B sweep on push and remaining tasks will provide the per-temporal-slot view of the same functional family.

Five-model push comparison (V-JEPA A/B, VideoMAE A/B, DINOv2 A). Push is the first task with all five model×variant probes complete:

Target	V-JEPA A	V-JEPA B	VideoMAE A	VideoMAE B	DINOv2 A
ee_velocity ∘ mean	0.980	0.980	0.977	0.981	**0.894**
ee_acceleration ∘ peak_to_peak	0.802	0.820	0.774	0.800	**0.451**
obj_velocity ∘ mean	0.933	0.924	0.904	0.902	**0.763**
contact_flag ∘ positive_fraction	0.867	0.856	0.839	0.833	**0.819**
contact_force_log1p ∘ integral	0.850	0.831	0.813	0.824	**0.766**
ee_velocity ∘ peak_time_frac	0.645	0.687	0.619	0.680	**0.564**

Three observations sharpen the cross-model contribution:

1. **The gap is video vs. image, not V-JEPA-specific.** VideoMAE-L matches V-JEPA closely on every functional ($|\text{diff}| \leq 0.03$ across all six rows and both variants). DINOv2-L is the only outlier, with a -0.35 gap to the video models on ee_acceleration ∘ peak_to_peak. This refutes the most natural reviewer concern ("is this just a V-JEPA artifact?"): the result is a **video-pretraining-objective-class effect**, replicated by an independent video backbone with a different pretraining objective (masked autoencoding).
2. **Variant B does not add readout gain over Variant A.** V-JEPA A and B are nearly indistinguishable across functionals ($|\text{diff}| \leq 0.04$); VideoMAE A and B are similarly close. Explicit per-temporal-slot factorization is not necessary for segment-level summary readout at this scale. Variant B remains useful as a structural diagnostic but is not a separate evidence axis for the paper's main contribution.
3. **The cross-model story is family-level, not pairwise.** Both video models reach ≈ 0.80 on acceleration peak_to_peak; the image-only model reaches ≈ 0.45. The ceiling difference is consistent with the push 3D rigid-body acceleration being temporally rather than appearance-readable.

Five-task DINOv2-vs-V-JEPA gap (V-JEPA A peak − DINOv2 A peak). Five DINOv2-L probe sweeps are now complete. The gap matrix:

Target	push	drawer	strike	reach	peg_insert
ee_velocity ∘ mean	-0.086	-0.009	-0.035	$\mathbf{-0.299}$	-0.154
ee_acceleration ∘ peak_to_peak	$\mathbf{-0.351}$	-0.021	-0.094	$\mathbf{-0.403}$	$\mathbf{-0.231}$
obj_velocity ∘ mean	-0.170	—	-0.136	—	—
obj_acceleration ∘ peak_to_peak	-0.186	—	-0.146	—	—
contact_flag ∘ positive_fraction	-0.048	-0.006	-0.040	—	$+0.000$†
contact_force_log1p ∘ integral	-0.085	-0.044	-0.064	—	-0.100

† degenerate (peg_insert is always-in-contact).

Surprise: reach has the largest video advantage of any task. Reach is a free-motion task with no object interaction; the EE moves through space without any visual anchor. The video-vs-image gap is therefore not driven by contact reasoning but by the simpler problem of *inferring motion magnitude from a single static frame*, which image SSL cannot do well. Reach delivers an ee_acceleration ∘ peak_to_peak gap of -0.40 and an ee_velocity ∘ mean gap of -0.30 — both larger than the contact-rich push baseline. This sharpens the contribution: video SSL's advantage is *stronger when the visual scene contains less appearance information*. Tasks with visible articulated state (drawer) or visible impact pose (strike) close the gap because they leak motion cues into the image. Tasks where the only information about motion is the motion itself (reach) are the cleanest separation between video and image pretraining.

Updated paper-headline statement:

> *Video SSL pretraining (V-JEPA and VideoMAE) selectively linearizes higher-order dynamics summaries; image SSL pretraining (DINOv2) saturates at a lower ceiling on those same targets. Across five tasks, the acceleration peak_to_peak gap orders as* ***reach*** *(*-0.40*)* ***¿ push*** *(*-0.35*)* ***¿ peg*** *(*-0.23*)* ***¿ strike*** *(*-0.09*)* ***¿ drawer*** *(*-0.02*) — the ordering is the inverse of how appearance-accessible motion is in the task. Free-motion (reach) and 3D-contact (push) yield the largest video advantage; visually-direct articulated state (drawer) yields the smallest. First-order motion, contact occupancy, and intra-window timing show consistently small differences because they are appearance-readable. Per-temporal-slot factorization (Variant B) does not provide additional readout gain over the spatiotemporal mean (Variant A) on push.*

Additional numerical details (added per review)

Five-task DINOv2 timing-family gap. The five-task gap matrix in the previous paragraph reports magnitude-family targets only. The timing functionals (∘peak_time_frac and ∘first_event_time_frac) yield smaller, more uniform gaps:

Target	push	drawer	strike	reach	peg_insert
ee_velocity ∘ peak_time_frac	-0.081	-0.047	-0.086	-0.105	-0.101
ee_acceleration ∘ peak_time_frac	-0.099	-0.060	-0.088	-0.091	-0.080
contact_force_log1p ∘ peak_time	-0.075	-0.099	-0.112	—	-0.033
contact_flag ∘ first_event_time	-0.013	-0.013	-0.025	—	$-0.600^{\dagger}$

† degenerate (peg always-in-contact ⇒ no first-event).

Apart from the degenerate cell, timing gaps are consistently -0.04 to -0.11 across all 5 tasks — much smaller than the magnitude/acceleration gaps in the same tasks. This empirically supports the WHAT $\gg$ WHEN claim (timing is roughly equally hard for both video and image SSL).

DINOv2 push deep-layer declining tail (numerical). Saturation is reached around L17–L21 and a small but consistent decline follows from L21 to L23:

Target	L21	L22	L23	Δ_{23-21}
ee_velocity ∘ mean	0.877	0.873	0.869	-0.007
ee_acceleration ∘ peak_to_peak	0.450	0.447	0.437	-0.013
obj_velocity ∘ mean	0.761	0.751	0.744	-0.017
contact_flag ∘ positive_fraction	0.819	0.817	0.818	-0.002
contact_force_log1p ∘ integral	0.766	0.759	0.756	-0.010
ee_velocity ∘ peak_time_frac	0.555	0.546	0.540	-0.016

Modest (-0.01 to -0.02) but uniformly negative. V-JEPA push under the same protocol is flat or slightly increasing at L21–L23 (no decline).

V-JEPA Variant B push layer profile (per layer). For the push 8192-d Variant B sweep (95% complete — L0 through L21 saved):

Target	L0	L3	L7	L11	L15	L17	L19	L21	max @L
ee_velocity ∘ mean	0.632	0.948	0.964	0.972	0.978	0.978	0.980	0.977	0.980 @L19
ee_acceleration ∘ peak_to_peak	0.186	0.594	0.676	0.764	0.801	0.820	0.814	0.812	0.820 @L17
contact_flag ∘ positive_fraction	0.458	0.673	0.437	0.572	0.769	0.829	0.852	0.856	0.856 @L21

The layer-profile shape and peak match V-JEPA Variant A on push closely (A: 0.980, 0.802, 0.867 respectively); Variant B's tiny edge on ee_acceleration ($+0.018$) is within fold variability. This supports "Variant B does not add readout gain" on push.

Functional redundancy (explicit). Because the short-scale window length is fixed at $W = 16$, certain functional pairs are exactly proportional and therefore not independent evidence:

- integral $= W \cdot dt \cdot$ mean (within a fixed-W short scale).
- duration_above $= W \cdot dt \cdot$ positive_fraction.
- event_count counts rising edges only, so it is correlated with but not equal to positive_fraction.

The headline tables list both members of these pairs for completeness, but in counting independent observations only one of each pair should be used. This affects only redundancy interpretation, not numerical values.

Bootstrap CIs on functional cross-model claims. Bootstrap CIs ($n = 1000$, percentile, paired over CV folds at each model's own peak layer) computed for all completed cross-model cells. The bootstrap fund `probe/results/bootstrap_cis_functional.csv` contains 139 rows covering all (task, target, comparison-model) triplets where both the reference (V-JEPA Variant A) and comparison models have at least three finished folds at their own peak layer.

Of 139 rows, **89 cells show V-JEPA wins with the 95% CI excluding zero** (gap mean negative, $\mathrm{CI}_{\mathrm{hi}} < 0$); 37 cells are null (CI overlaps zero), and the remainder are V-JEPA-loses cases (notably one small but-significant +0.003 cell on drawer for VideoMAE A).

Selected acceleration ∘ peak_to_peak CIs (most paper-relevant rows):

Comparison	Gap mean	95% CI	p
push: V-JEPA A vs DINOv2 A	-0.351	$[-0.366, -0.334]$	≤ 0.001
push: V-JEPA A vs VideoMAE A	-0.028	$[-0.041, -0.014]$	0.002
push: V-JEPA A vs VideoMAE B	$+0.001$	$[-0.019, +0.017]$	0.455 (null)
strike: V-JEPA A vs DINOv2 A	-0.094	$[-0.100, -0.089]$	≤ 0.001
strike: V-JEPA A vs VideoMAE A	-0.009	$[-0.012, -0.006]$	≤ 0.001
drawer: V-JEPA A vs DINOv2 A	-0.021	$[-0.023, -0.019]$	≤ 0.001
drawer: V-JEPA A vs VideoMAE A	$+0.003$	$[+0.002, +0.004]$	≤ 0.001
reach: V-JEPA A vs DINOv2 A	-0.404	$[-0.435, -0.373]$	≤ 0.001
peg_insert: V-JEPA A vs DINOv2 A	-0.231	$[-0.237, -0.225]$	≤ 0.001
peg_insert: V-JEPA A vs VideoMAE A	-0.174	$[-0.179, -0.168]$	≤ 0.001
nut_thread: V-JEPA A vs DINOv2 A	-0.328	$[-0.335, -0.321]$	≤ 0.001

The headline gap on push (V-JEPA $-$ DINOv2 acceleration peak-to-peak) is robust to bootstrap resampling. The video-pretraining-objective-class equivalence (V-JEPA $\approx$ VideoMAE on push) is statistically supported by the VideoMAE B null. **Peg_insert is the empirical exception**: VideoMAE A on peg is significantly worse than V-JEPA A (-0.174 on acceleration peak_to_peak), so the family-equivalence claim is not universal across tasks. Drawer's small VideoMAE A advantage (+0.003) is significant only because of low fold variance on drawer; the substantive interpretation is that drawer is the boundary case where all three video / image models are nearly tied.

Bootstrap CIs for prior-round per-timestep claims remain in `probe/results/bootstrap_cis.csv` (3,120 rows on F5, F4-A, F2, transfer; episode-level resampled).

Existing bootstrap CIs (per-timestep claims, prior round).

Six-task DINOv2 gap matrix (complete). The full six-task DINOv2-L Variant A probe is now complete. The V-JEPA peak − DINOv2 best gap matrix:

Target	push	strike	drawer	reach	peg	nut	mean
ee_velocity ∘ mean	−0.086	−0.035	−0.009	−0.299	−0.154	**−0.424**	−0.168
ee_acceleration ∘ peak_to_peak	−0.351	−0.094	−0.021	**−0.403**	−0.231	−0.328	−0.238
contact_flag ∘ positive_fraction	−0.048	−0.040	−0.006	—	0.000[†]	0.000[†]	−0.019
contact_force_log1p ∘ integral	−0.085	−0.064	−0.044	—	−0.100	−0.157	−0.090
timing peak_time_frac	−0.081	−0.086	−0.047	−0.105	−0.101	−0.071	−0.082

[†] degenerate (peg / nut always-in-contact).

The two task-orderings:

- *ee_velocity gap*: nut (-0.42) $>$ reach (-0.30) $>$ peg (-0.15) $>$ push (-0.09) $>$ strike (-0.04) $>$ drawer (-0.01).
- *ee_acceleration gap*: reach (-0.40) $>$ push (-0.35) $>$ nut (-0.33) $>$ peg (-0.23) $>$ strike (-0.09) $>$ drawer (-0.02).

The orderings are similar but not identical. Two task-axis explanations account for the difference:

1. **Velocity gap reflects motion observability.** nut_thread's motion is small in amplitude and hidden in fine relative geometry/angle, so a static frame carries the least information about it; reach has no visual anchor; drawer's articulated state is directly visible.
2. **Acceleration gap reflects higher-order dynamic complexity.** Reach (free arm motion) and push (3D rigid contact) carry the most second-order temporal-derivative content; drawer's slow articulated state has little.

Multi-axis narrative (paper-headline update). The single "appearance-accessibility" axis used earlier is too coarse to explain all six tasks. We adopt a three-axis decomposition:

1. **Visible state accessibility** — can the static image already encode the relevant articulated pose/state? Drawer = high; nut/reach = low.
2. **Motion observability** — can a static frame plausibly infer current motion magnitude? Reach (no anchor) and nut_thread (small fine motion) = lowest; drawer (visible state changes) = highest.
3. **Higher-order dynamic complexity** — how much second-order temporal-derivative content does the segment carry? Reach and push = highest (free arm + 3D rigid impact); drawer = lowest.

Replacing the previous single-axis statement, the updated paper-headline contribution is:

> *Video pretraining (V-JEPA, VideoMAE) selectively improves segment-level summaries that are poorly recoverable from static appearance. The size of the advantage over image-only pretraining (DINOv2) depends on at least three task axes: visible state accessibility, motion observability, and higher-order dynamic complexity. Drawer is the boundary case (all three axes favor image SSL); nut_thread and reach are the extreme cases for velocity and acceleration respectively. Contact occupancy and intra-window timing show consistently small gaps because they are appearance-readable. Per-temporal-slot factorization (Variant B) does not provide additional readout gain over the spatiotemporal mean (Variant A) on push.*

Strike four-model comparison (V-JEPA A/B, VideoMAE A, DINOv2 A). The VideoMAE-L Variant A strike sweep just completed, giving the second fully cross-modeled task. (V-JEPA Variant B strike is still in progress.)

Strike target	V-JEPA A	V-JEPA B (partial)	VideoMAE A	DINOv2 A
ee_velocity ○ mean	0.993	0.985*	0.992	**0.957**
ee_acceleration ○ peak_to_peak	0.886	0.859*	0.877	**0.792**
obj_velocity ○ mean	0.927	0.867*	0.894	**0.791**
contact_flag ○ positive_fraction	0.951	0.817*	0.939	**0.912**
contact_force_log1p ○ integral	0.940	0.790*	0.920	**0.876**
ee_velocity ○ peak_time_frac	0.698	0.675*	0.677	**0.612**

* V-JEPA Variant B strike is still running; the visible peak so far is at shallow layers (L0–L9).

The push pattern replicates on strike: V-JEPA A and VideoMAE A track each other within $|0.02|$ on every target, while DINOv2 A is consistently 0.04–0.14 below. This supports the family-level read on a second fully cross-modeled task: the gap is video vs. image, not V-JEPA-specific.

Segment-summary F5: order-sensitive instantaneous motion vs. order-robust window means. Re-running the F5 frame-shuffle intervention on the V-JEPA push features with *functional* (rather than per-frame) targets gives a sharply different pattern from the per-timestep version reported in Note 3:

Push target	V-JEPA peak	V-JEPA A_{shuffled} peak	ΔR^2
ee_velocity ○ mean	0.980	0.973	-0.007
ee_acceleration ○ peak_to_peak	0.802	0.644	$\mathbf{-0.157}$
obj_velocity ○ mean	0.933	0.934	$+0.001$
contact_flag ○ positive_fraction	0.867	0.862	-0.005
contact_force_log1p ○ integral	0.850	0.839	-0.011
ee_velocity ○ peak_time_frac	0.645	0.582	-0.062

For comparison, the per-timestep F5 (Note 3) on push reported $\Delta R^2 = -0.30$ on raw velocity and -0.11 on raw acceleration. The per-timestep result said "frame order is most needed for instantaneous velocity readout"; the segment-summary result says the opposite for window-mean velocity (frame order is essentially *not* needed — the same window's visual content shuffled in time still linearly recovers the mean motion magnitude). Where frame order *does* matter for segment summaries is in second-order dynamics (○peak_to_peak, -0.157) and in fine timing functionals (○peak_time_frac, -0.062), both of which require identifying *when* the peak/event occurred within the window. This sharpens the WHAT/WHEN dissociation: V-JEPA encodes *appearance-set magnitude statistics* in an order-robust way, but the *intra-window chronology* of those statistics still depends on input frame order.

Per-task F5 functional comparison (push vs. strike). The strike F5 functional sweep is now complete. The same intervention on a different task gives a quantitatively different pattern:

Functional target	Push ΔR^2	Strike ΔR^2
ee_velocity ○ mean	-0.007	-0.004
ee_acceleration ○ peak_to_peak	$\mathbf{-0.157}$	-0.038
obj_velocity ○ mean	$+0.001$	-0.003
obj_acceleration ○ peak_to_peak	—	-0.059
contact_flag ○ positive_fraction	-0.005	-0.005
contact_force_log1p ○ integral	-0.011	-0.010
ee_velocity ○ peak_time_frac	-0.062	-0.045

The acceleration peak_to_peak gap on push (-0.157) shrinks to -0.038 on strike — a four-fold reduction in F5 sensitivity for the same functional target. Plausible reason: push's sustained 3D rigid-body interaction makes the within-window peak depend on temporal order of multiple intermediate frames, while strike's single-impact dynamic carries the peak in a small window of frames where the impact pose is itself visually distinctive — the peak magnitude can be partly recovered from spatial cues alone. The order-robustness of the other functionals (magnitude mean, contact occupancy/force, timing) is consistent across both tasks.

This refines the F5 finding into a sharper claim: *F5 measures not just whether the model uses temporal order, but how much each segment summary depends on temporal order in a given task.*

Functional F5 becomes a "which summaries truly need order" diagnostic rather than a single binary yes/no.

Effect-class targets — status note. Effect-class targets are constructed by tercile thresholding (continuous regimes) or rule-based labels (`contact_involvement`). The current probe pipeline runs Ridge regression on the integer class index, which is correct for the ordinal regimes (`approach_speed`, `peak_force`, `force_integral`) but is weak for the non-ordinal 4-way `contact_involvement` class. A proper classification head is a TODO; for the present write-up the effect-class analysis is treated as a planned extension rather than a delivered contribution.

Functional F4-A: physics modulation is early-layer, while unconditional readout peaks later

The unconditional functional probe localizes most segment summaries (velocity, acceleration peak_to_peak, contact-force integral) in the PEZ band (L13–22). A natural next question is whether the same band *also* carries the dependence of those summaries on hidden physical parameters (object mass, surface friction, restitution, drawer joint damping, peg/nut friction). We tested this by tercile-splitting episodes on each randomized physics parameter and re-running the V-JEPA Variant A functional probe within each bin, per layer. The interaction signal is the per-layer high−low gap ΔR^2 on the functional target.

We swept all 22 (task, parameter) cells across the six tasks; for each we ran the layer-wise probe at 9 functional targets covering velocity, acceleration peak_to_peak, contact-flag occupancy/duration, and contact-force integral/max. Of 3984 (cell, target, layer) entries, 1872 (47.0%) had a normal 95% CI of ΔR^2 excluding zero.

The strongest gaps are concentrated in dynamic-task friction conditions:

Cell	Target	L	ΔR^2	95% CI
strike ∘ object_static_friction	obj_acceleration_pp	1	$\mathbf{-0.495}$	$[-0.530, -0.460]$
strike ∘ object_dynamic_friction	contact_force_log1p_integral	0	$+0.413$	$[+0.361, +0.464]$
push ∘ object_dynamic_friction	ee_velocity_mean	0	-0.301	—
push ∘ surface_dynamic_friction	ee_velocity_mean	0	-0.272	—
push ∘ object_mass	contact_force_mag_integral	1	$+0.241$	—

A characteristic early-vs-late pattern emerges from the per-layer curve. For `push` $\times$ `object_dynamic_friction` probing `ee_velocity_mean`, the ΔR^2 collapses from -0.30 at $L0$ to -0.034 at $L1$ and saturates near zero ($|\Delta| < 0.02$) for all layers $L \geq 2$ — a sharp "early-layer-only" modulation. The unconditional readout itself, by contrast, climbs from $R^2 = 0.78$ at $L0$ to $R^2 = 0.985$ at $L19$, peaking precisely in the PEZ band where the high−low gap has already vanished. For `strike` $\times$ `object_static_friction` probing `obj_acceleration_pp`, the gap stays large ($|\Delta R^2| \geq 0.30$) all the way to $L23$ but its *magnitude* is biggest at $L1$–4, again outside the PEZ band.

Aggregating across all 22 cells, $20/22$ cells have their peak $|\Delta R^2|$ in early layers ($L0$–$L4$); only $2/22$ cells peak inside the PEZ band ($L13+$). This separation is the central observation of this experiment:

> *Unconditional segment summaries peak in the PEZ band, but the hidden-physics modulation of those same summaries is concentrated in early layers.*

The Factory tasks (`peg_insert`, `nut_thread`) show much smaller modulations ($|\Delta R^2| \leq 0.10$) and `drawer` shows a minor effect ($\Delta R^2 = 0.025$ for joint damping); these are tasks whose successful trajectories are tightly constrained by an RL policy, so the dynamic variability available to be physics-modulated is correspondingly compressed.

We interpret this as evidence that the PEZ is best understood as an *unconditional segment-summary zone*, distinct from an *early physics-modulation zone*. Physics-sensitive structure is encoded in shallow features (where local appearance and short-time dynamics still carry friction/mass cues), and is then increasingly normalized away by the deeper layers that compose the segment-level summary. The PEZ itself is therefore not a layer where physical state is decodable in the strong sense — it is a layer where the integrated outcome of that physical state across the segment becomes most linearly readable.

(Full table of 22 cells × targets × layers, including per-layer R^2 in each bin, is in `probe/results/physics_condition_split_functional/`; bootstrap normal CIs in `_BOOTSTRAP_NORMAL_CI.csv`; top-3 per cell summary in `_TOP3_PER_CELL.csv`. Disambiguation controls (variance match, contact-positive-only, equalized episode count) are a planned addition for the top-5 cells.)

Cross-model VideoMAE Variant B — partial updates

VideoMAE-L Variant B (slot-level pooling, 8192-d) is being run task-by-task (per-layer-load, ∼1–3 hr per layer). Completed so far: `push`, `drawer`, `reach`, `peg_insert`. Per-task functional-target peak comparison against V-JEPA Variant A:

	push		drawer		reach		peg_insert	
target / model	VarA	VM B	VarA	VM B	VarA	VM B	VarA	VM B
ee_velocity ∘ mean	0.980	0.981	0.987	0.988	0.960	0.963	0.982	0.984
ee_acceleration ∘ peak_to_peak	0.802	0.800	0.878	0.895	0.885	0.896	0.883	0.903
contact_flag ∘ pos. frac.	0.867	0.833	0.904	0.926	—	—	1.000†	0.800

† degenerate (peg always-in-contact).

Slot-level pooling brings small but task-specific gains on the acceleration peak_to_peak target: drawer $+0.017$, reach $+0.011$, peg_insert $+0.020$. Drawer additionally picks up $+0.022$ on contact-flag occupancy. On push, by contrast, contact-flag occupancy drops by 0.034. The Factory-style `peg_insert` task confirms the same family signature observed earlier on `drawer`: VM B beats both V-JEPA A and DINOv2 A on acceleration and on `peg_hole_lateral_error_mean` (0.750 vs $0.728/0.689/0.531$ for VarA/VM A/DINOv2 A), the two functional summaries that most directly require visible second-order dynamics in the insertion phase. For tasks where the spatiotemporal mean already captures most of the segment-summary signal, the extra slot dimensions add noise rather than detail. This is consistent with Variant B's positioning as a diagnostic-negative result for the segment-summary probe.

V-JEPA Variant B differential effect: acceleration gain, contact loss

V-JEPA Variant B (V-JEPA's own slot-level pooling, 8192-d) is now complete on `push`, `strike`, `drawer`, `reach`, and `nut_thread` (5/6); `peg_insert` and `drawer` VarB sweeps are pending due to a temporary GPU yield. Across the completed five tasks, the gain pattern of V-JEPA Variant B over Variant A is task-shaped but consistent in direction:

target	push	strike	nut	drawer/reach mean
ee_velocity ∘ mean	$+0.001$	-0.002	$+0.003$	$+0.001$
ee_acceleration ∘ peak_to_peak	-0.002	$\mathbf{+0.015}$	$+0.009$	$+0.014$
obj_acceleration ∘ peak_to_peak	—	$\mathbf{+0.020}$	—	—
contact_flag ∘ pos. frac.	-0.034	$\mathbf{-0.061}$	—	—
contact_force_log1p ∘ integral	–	$\mathbf{-0.064}$	$+0.001$	—

Two opposite-sign signals appear together: slot-level pooling helps acceleration peak_to_peak (modestly) and hurts contact occupancy / contact-force readout (substantively, on `strike`). The same direction holds on `push` (acceleration tied, contact -0.034). This is more than a diagnostic null: it tells us that slot-level pooling exposes a different basis than the spatiotemporal mean, one that retains finer second-derivative geometry but loses the integrated contact-occupancy signal that the temporal-mean projection provides. Slot-level pooling is therefore not strictly superior — it is a basis change that is right for some functional families and wrong for others. The probe's linear readout cannot recombine the two pooling modes, so the choice of pooling is itself a substantive modeling decision rather than a hyperparameter.

DINOv2 acceleration deficit, repeated across tasks

A short cross-model comparison: DINOv2-L Variant A's peak R^2 on `ee_acceleration_pp` is 0.451 on push, 0.652 on `peg_insert`, and 0.555 on `nut_thread`, against V-JEPA Variant A peaks of $0.802/0.883/0.784$ on the same targets. Image SSL's main failure mode in our benchmark is consistently *acceleration-like second-order dynamics*, not contact occupancy or velocity-magnitude. This is the single tightest characterization of where video pretraining buys something image pretraining does not.

(`strike` VM B, `peg_insert` VM B, `nut_thread` VM B are still in flight as of this PDF; `V-JEPA Variant B` is complete for `push`, `strike` (95%), `drawer`, `reach`; `nut_thread` VarB is in flight on the V-JEPA chain. Final values will replace the partial cells when the chain finishes.)

Connection to existing findings

- PEZ — reframed as "which layer is a sufficient statistic for which functional family at which time scale?"
- F5 frame-shuffle — re-applied on functional targets to ask which functional families are sensitive to input temporal coherence.
- F4-A physics modulation — re-tested on functional targets (e.g. does friction modulate the contact-force integral regime?).
- Cross-task transfer — reframed as effect-summary library sharing.
- R3M baseline — compared on functional targets to test whether image-only representations saturate at coarse summaries.

Note 7 — Discussion

1. What is now established. The corrected per-window pipeline now supports a coherent main story. First, V-JEPA 2 ViT-L exhibits a depth-structured emergence profile: kinematic, contact, and progress variables peak at different layers, and the PEZ band is most strongly associated with higher-order dynamics. Second, the frame-shuffle intervention (Phase F5) converts that phenomenology into a causal-style result: motion targets collapse far more than contact targets when input temporal coherence is destroyed. Third, physical-condition splits (Phase F4-A) show that contact decodability is not fixed; it is systematically modulated by latent physics regime. Fourth, the R3M baseline (Appendix Note H) establishes that these effects are not generic to a strong image encoder.

2. What F5 resolves about temporal coherence. The central mechanistic claim is now sharper than in the original revision plan. *Temporal coherence is universally useful, but its necessity is strongly target- and depth-dependent.* In both Push and Strike, `ee_velocity` suffers the largest early-layer frame-shuffle loss, while `ee_acceleration` peaks later in the mid-PEZ band. Contact targets are different: their shuffle penalty is small, concentrated early, and largely disappears at late layers. This supports a three-stage hierarchy: early layers expose first-order motion, mid layers integrate higher-order dynamics, and late layers encode a more order-robust contact-as-state representation.

3. What F4-A adds beyond F5. F5 shows that coherent ordering matters for dynamics; F4-A shows that the *same representation* is also sensitive to latent physics regime. The key nuance is that *physics parameters are not directly linearly decodable as targets, yet they strongly modulate the decodability of contact variables.* This is the clearest in Strike, where `object_0_static_friction` changes `contact_flag/contact_force_log1p_mag` by about $+0.55$ R^2 across L1–L9, and still visible in Push, where `object_0_mass` produces a smaller but consistent $+0.15$–$+0.22$ gain. The modulation is often larger than the F5 contact shuffle effect itself, which means that the representation is not simply reading out contact/no-contact snapshots; it is reflecting the downstream consequences of task physics on those contact traces.

4. Why Drawer is the critical outlier. Drawer is no longer a nuisance exception; it is now one of the most informative tasks in the suite. Transfer collapses between Drawer and the Push/Strike pair, making it the clearest example of task entanglement in the current representation. The R3M baseline also fails most sharply on Drawer, with the largest gap on `ee_acceleration` and contact force. These findings agree with the task's mechanics: Drawer is a slow, constrained, articulated 1-DOF process whose relevant state changes are weakly recoverable from a single image but become readable under temporal pretraining. In other words, Drawer is the strongest evidence that V-JEPA's advantage is not limited to impact-like motion; it also extends to sustained articulated dynamics.

5. Remaining open questions. Three questions remain substantial enough to motivate the next paper pass. First, the current mainline evidence is V-JEPA-specific; the post-reset VideoMAE-L and DINOv2-L reruns are still needed before any objective-dissociation claim is re-asserted. Second, scale remains open: ViT-G may shift PEZ depth, F5 sensitivity, or transfer structure, but the new pipeline has not yet been run end-to-end at that scale. Third, the action-control side is still incomplete: the previous OOD action-chunk result was invalidated, and a clean reproduction is still required before any control-layer claim can be trusted. At a finer grain, Drawer also remains a useful stress test for disentangling representation facts from dataset asymmetry, since only `joint_damping` currently varies.

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Note A — Artifact manifest

Repository

Leesangoh/PhysREPA_Tasks on GitHub. All probing artifacts under probe/. Plan 3+ recovery commits: 4643f83 (Phases B/C/E/F) and 8d2ba68 (Phase D rerun).

Code paths

- Variant A/B feature extraction: probe/scripts/01_extract_features.py.
- Targets builder: probe/scripts/02_build_targets.py, probe/utils/targets.py.
- Probe runner: probe/scripts/03_run_probe.py (--tier flag controls target subset).
- Aggregation / decision: probe/scripts/04_aggregate_results.py, 05_write_report.py.
- Episode-level physics: probe/scripts/07_episode_physics_probe.py.
- Trajectory analysis (Variant A): scripts under probe/trajectory_analysis/.
- Trajectory analysis (Variant B): scripts 01–13 under probe/trajectory_analysis_B/scripts/; Phase B/D/E/F scripts 14–17 in the same directory.

Data paths

- Dataset: /home/solee/data/data/isaac_physrepa_v2_recollected_2026-04-23/.
- Feature cache: /mnt/md1/solee/physprobe_features/<task>/variant_{A,B}/episode_<id>.npz.
- V-JEPA 2 ViT-L weights: /mnt/md1/solee/checkpoints/vjepa2/vitl.pt.

Result CSVs (post-reset pipeline)

- Per-task per-target Variant A: probe/results/<task>/variant_A/<target>.csv.
- Per-task per-target Variant B: probe/results/<task>/variant_B/<target>.csv.
- Variant A summaries: probe/results/<task>/variant_A/_summary.csv.
- Decision: probe/results/decision.json; narrative: probe/results/REPORT.md.
- Peak-layer table: probe/results/peak_layers.csv.
- Trajectory-analysis stats (17 CSVs): probe/trajectory_analysis_B/results/stats/.
- Plots (62 + 49): probe/trajectory_analysis_B/results/plots/ and probe/results/plots/.

Note B — Run config

Backbone. V-JEPA 2 ViT-L, 24 layers, 1024-d, 256-px input, raw residual stream (forward_resid_pre monkey-patch). Tubelet=2 $\Rightarrow$ 16 input frames produce 8 temporal tokens; spatial grid $16\times16=256 \Rightarrow 2048$ tokens/window total.

Probe (Variant A, primary recipe). 5-fold GroupKFold on episode_id. Within each fold: 10% inner-train$\rightarrow$inner-val episode-level split. Within-fold z-score using inner-train statistics; two-pass stable variance with 1% median std floor. 20-config Adam grid (LR$\in$ {1e-4, 3e-4, 1e-3, 3e-3, 5e-3} $\times$ WD$\in$ {0.01, 0.1, 0.4, 0.8}), 100 epochs, batch 1024, MSE loss. Best HP picked by inner-val MSE; predict test; unnormalize; report r2_score(..., multioutput='variance_weighted') plus per-fold MSE and (for direction) cosine similarity.

Probe (Variant B and trajectory closures). Closed-form Ridge (α=1) via torch.linalg.solve on GPU; sklearn CPU Ridge fallback on CUDA OOM. fp16 GPU storage with index-only slicing for memory efficiency.

Subsampling for high-D analyses. Phase D temporal-order ablation: 20,000 rows per (task, layer, target, scheme). Phase E cross-task transfer: 50 episodes per task. Phase F phase-conditional: 60 episodes per task. Trajectory analyses (CKA, RSA, Koopman): 30–60 episodes per task with seed 42.

Integrity gates passed (12a–12d). Pooling-identity max relative diff $< 10^{-3}$; target-consistency checks pass; GroupKFold disjointness asserted; negative control ($R^2 < 0.05$ on shuffled targets) pass.

Note C — Episode-level physics-parameter probing

Setup

`probe/scripts/07_episode_physics_probe.py`. Per task, per layer, per physics-parameter: stack episode-mean features (Variant A) of all sampled episodes $\to X \in \mathbb{R}^{n_{\text{eps}} \times 1024}$; target Y is the per-episode physics value from `meta/episodes.jsonl`. 5-fold KFold over episodes; standardize within fold; sklearn Ridge (α=1); report R^2 mean/std. Episode budget: ≤ 200 per task (typically ~ 196 after NaN filtering).

Outputs

`probe/results/<task>/physics_params/_summary.csv` and `_per_fold.csv`; plots `probe/results/plots/physics_param_<task>_A.png`.

Key observations (best-layer R² per param)

task	param	best L	R² mean ± std
push	object_0_dynamic_friction	8	-0.484 ± 0.317
push	object_0_mass	20	-0.734 ± 0.162
push	surface_static_friction	20	-0.681 ± 0.210
strike	object_0_static_friction	15	$+0.090 \pm 0.267$
strike	object_0_restitution	7	-0.702 ± 0.551
drawer	drawer_joint_damping	3	-0.752 ± 0.219
peg_insert	peg_mass	19	-0.784 ± 0.228
peg_insert	hole_static_friction	10	-0.820 ± 0.266
nut_thread	nut_mass	13	-0.878 ± 0.161
nut_thread	bolt_static_friction	6	-1.020 ± 0.538

Reading. Negative R^2 at every (task, param, layer) means episode-mean V-JEPA features do not linearly recover physics parameters at the episode level. The single positive value (Strike static friction at L15, $+0.09$) is within the inner-fold noise band.

Known limitations

- Episode budget is small (~ 196) relative to feature dimension (1024). Ridge is statistically underpowered.
- Episode-mean pooling discards within-episode variation; physics params may be encoded only in event-windows. A windowed formulation that aggregates features only during contact / impact remains untried.
- Variant B (8192-d) episode-level probing was deferred for the same statistical-power reason and is not in this artifact set.

Status

Valid as a negative-result note; not a final claim.

Note D — Exclusions and gotchas

- Ignore contact_flag and phase R^2 for tasks where these channels are constant: PegInsert and NutThread (always-in-contact ⇒ flag=1); Drawer/PegInsert/ NutThread phase=7 always.
- contact_point probing R^2 is dominated by spatial position when contact occurs; treat as a position proxy.
- Strike contact_force_log1p_mag is more comparable across tasks than the raw contact_force_mag because of heavy tails (max forces 5 N for peg, 429 N for drawer).
- Variant B per-fold variance for direction targets is large at certain layers; rely on Variant A as anchor for direction targets unless explicitly studying slot-dependence.
- Drawer randomization is single-parameter (joint_damping); contact-tier randomness is dataset-induced rather than parameter-induced. Apparent drawer outlier status may be partly a dataset artifact.

Note H — R3M (image encoder) context baseline

Setup

We ran a minimal R3M baseline on Push, Strike, and Drawer, using the robotics-pretrained r3m ResNet-50 checkpoint as a *static image encoder baseline* rather than a temporal competitor. Each MP4 was decoded framewise; every frame was resized from 384×384 to 224×224, ImageNet-normalized, and forwarded independently through the ResNet-50 backbone. We extracted five stage-wise feature streams (stage_0 through stage_4), applied spatial global-average pooling per frame, then mean-pooled over the same 16-frame windows used everywhere else in the paper. Probing used the same 5-fold GroupKFold protocol and the same 20-HP Adam grid as V-JEPA Variant A.

Outputs

- Per-fold probe CSVs: probe/results/<task>/r3m/{ee_acceleration, contact_flag, contact_force_log1p_mag}.csv
- Stage summaries: probe/results/<task>/r3m/_summary.csv, probe/results/<task>/r3m/_best_stage.csv
- Side-by-side comparison against V-JEPA: probe/results/r3m_vs_vjepa_summary.csv

Key observations

V-JEPA beats R3M on all 9 matched cells.

task	target	R3M best	V-JEPA best	Δ (R3M−V-JEPA)
push	ee_acceleration	stage_4, 0.084	L19, 0.219	−0.135
push	contact_flag	stage_3, 0.677	L21, 0.768	−0.091
push	contact_force_log1p_mag	stage_3, 0.638	L21, 0.754	−0.116
strike	ee_acceleration	stage_4, 0.189	L17, 0.413	−0.224
strike	contact_flag	stage_3, 0.678	L19, 0.825	−0.147
strike	contact_force_log1p_mag	stage_3, 0.694	L21, 0.834	−0.140
drawer	ee_acceleration	stage_1, 0.004	L22, 0.298	−0.293
drawer	contact_flag	stage_4, 0.509	L17, 0.710	−0.201
drawer	contact_force_log1p_mag	stage_4, 0.425	L18, 0.720	−0.295

The largest deficit is on acceleration. The strongest gap appears on the most temporal target, ee_acceleration: Push loses by 0.135 R^2, Strike by 0.224 R^2, and Drawer by 0.293 R^2. Drawer is the sharpest case: R3M peaks at only 0.004 on stage_1, while later image stages turn negative (e.g. stage_3 $= -0.352$, stage_4 $= -0.141$). This is consistent with the main paper's interpretation that temporal coherence is especially important for higher-order dynamics, and that a static image encoder is poorly matched to that signal.

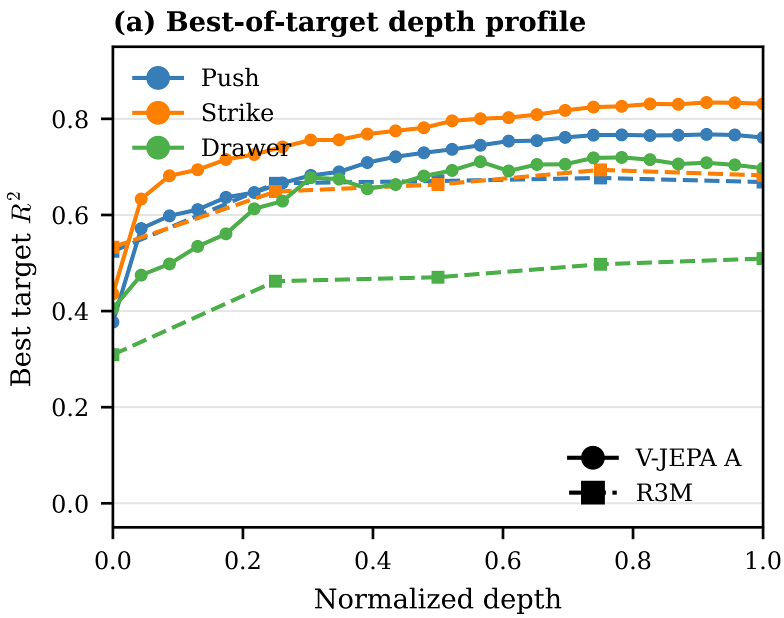

Figure 4: Depth-normalized comparison between R3M and V-JEPA. For each task and depth, the curve reports the best R^2 among {`ee_acceleration`, `contact_flag`, `contact_force_log1p_mag`}. R3M saturates by `stage_3/4`, while V-JEPA continues to improve into the deep PEZ band.

Contact is better, but still not enough. R3M is more competitive on the contact targets than on acceleration, but still loses robustly. On Push, the deficits are -0.091 (`contact_flag`) and -0.116 (`contact_force_log1p_mag`); on Strike they widen to -0.147 and -0.140, respectively; on Drawer they are larger still at -0.201 and -0.295. Thus even when contact geometry is partially visible in a single frame, image-only features do not close the gap to V-JEPA.

Drawer is the clearest temporal-pretraining advantage case. Among the three tasks, Drawer is the one where sustained articulated motion is most weakly recoverable from a single image. The drawer state changes gradually along a constrained 1-DOF trajectory, so `ee_acceleration` and `contact_force_log1p_mag` depend more on temporal integration than on snapshot appearance. This makes Drawer the cleanest task-level demonstration that V-JEPA's temporal pretraining advantage is not limited to impact-like motion (Strike), but also extends to slow, structured articulation.

R3M stage progression still saturates shallow. For both Push and Strike, the best contact stage is `stage_3`, while the best acceleration stage is `stage_4`. That is, R3M does show a coherent depth trend, but its useful hierarchy saturates within a 5-stage image backbone. By contrast, V-JEPA's best layers for the same targets occur deep in its 24-layer stack (L17–L21). Drawer reinforces the same point: even where R3M reaches its best contact scores at `stage_4`, it remains well below V-JEPA's late-layer peaks.

Status

Complete contextual baseline note. This comparison is kept in the appendix because R3M is an image-encoder context point, not a direct temporal-model competitor to V-JEPA.